

Optimization Study of SVM Classification in a Synthetic PA Cohort

SVM: Hard-Margin Feasibility, Soft Margins, and Kernel Validation

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Code and data: https://github.com/nttssv/convex_optimization_svm_example

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1 Problem Identification

Primary aldosteronism (PA) is an important cause of secondary hypertension. The clinically relevant subtype decision is whether PA is unilateral or bilateral. Unilateral PA can often be treated surgically, whereas bilateral PA is usually managed medically [3]. The aim of this course project is to formulate that binary classification task as a maximum-margin optimization problem, test whether a linear hard-margin support vector machine (SVM) is feasible on the supplied synthetic data, and, if it is not, identify a cross-validated soft-margin alternative. It is an optimization demonstration, not a clinical decision-support system or evidence of diagnostic utility.

The analysis begins with a *linear hard-margin* SVM to determine whether a perfect linear separator exists. That feasibility audit uses no slack variables, finite penalty parameter, hinge-loss penalty, or kernel transformation. Once the audit proves that the classes are not linearly separable, the report introduces a linear soft-margin SVM, derives its primal and dual formulations, and selects its penalty parameter C by leakage-safe grid search. The resulting sequence is therefore hard-margin feasibility, soft-margin optimization, nested cross-validated performance, and an RBF kernel comparison with explicit overfitting diagnostics.

2 Dataset and Feature Representation

The source report describes a cohort of 114 patients evaluated for PA at Changi General Hospital [4]. That internal report is cited for provenance but is not redistributed in the public repository. Patient-level clinical data were not available for this course project, so the numerical analysis uses a synthetic CGH-like cohort with 114 observations: 56 labeled unilateral PA (UPA) and 58 labeled bilateral PA (BPA). The synthetic data are used to test the optimization workflow and are not evidence of clinical performance.

Ten numerical predictors are used: plasma aldosterone concentration, plasma renin activity, potassium, tumor size, 18-hydroxycortisol, 18-oxocortisol, systolic blood pressure, diastolic blood pressure, antihypertensive burden measured in defined daily doses (DDD), and age. Age, DDD, and some laboratory variables are stored at integer precision, so “numerical” is more accurate than “continuous.” Let $x_i \in \mathbb{R}^p$ denote the feature vector for observation i , with $p = 10$, and let

$$t_i = \begin{cases} +1, & \text{for UPA,} \\ -1, & \text{for BPA.} \end{cases} \quad (2.1)$$

The four positively skewed hormone measurements are transformed as $\log(x + 10^{-6})$; the small guard is included for numerical safety although all generated assay values are positive. Within each cross-validation split, medians, means, and standard deviations are estimated from the training fold only; these training-fold quantities are then used for imputation and standardization of the held-out fold. No value is missing in the realized synthetic data, so median imputation is present for pipeline completeness but is inactive here.

2.1 Synthetic Data Generation and Scope

The synthetic cohort is generated reproducibly by `scripts/make_synthetic_dataset.py` with NumPy random seed 20260710. The subtype label is drawn first, after which the ten numerical predictors are sampled conditionally on that label. Age, blood pressure, antihypertensive burden, potassium, and tumor size are generated from clipped Gaussian distributions; PAC, PRA, 18-OHF, and 18-oxoF are generated from log-normal distributions. Prespecified class-conditional

shifts impose a clinically plausible simulated signal: UPA observations tend to have higher PAC, lower PRA and potassium, larger tumors, and higher hybrid-steroid values. The generated variables are then clipped to plausible ranges and rounded to the precision stored in the project CSV file. The realized sample contains 56 UPA and 58 BPA observations.

The generator was calibrated only to approximate selected marginal means from the source report. It does not reproduce patient-level records, a fitted multivariate covariance structure, or the full heterogeneity of a clinical population. Clipping and rounding also make realized sample means differ from their generating targets and can create boundary piles or ties. Consequently, both separability and predictive performance partly reflect the prespecified class-conditional shifts and simulation assumptions. Before clipping and rounding, each predictor is drawn conditionally independently given subtype, with a class-shifted mean and common within-class dispersion. After the stated hormone log transforms, the Bayes-optimal boundary is exactly linear up to clipping and rounding. The kernel comparison of Section 7 therefore has essentially no intentional nonlinear structure available to detect: it demonstrates the selection protocol rather than providing evidence about kernels for this clinical problem. A binary Sex variable is also generated for descriptive context but is excluded a priori from the SVM feature vector so that the analysis remains restricted to the ten numerical predictors listed above; the reported models are therefore not adjusted for sex.

Table 1 makes the calibration mismatch explicit. These are marginal targets, not constraints imposed exactly on the realized sample; they are shown to document effects of random sampling, clipping, and rounding rather than to validate the generator.

Table 1: Generator calibration targets from the source-summary values versus realized synthetic-cohort means. Diastolic blood pressure had no recorded target in the generator and is therefore omitted.

Predictor	Calibration target	Realized mean	Difference
Age	48.00	49.36	+1.36
Systolic BP	141.00	139.42	-1.58
DDD	2.39	2.31	-0.08
Potassium	3.59	3.59	-0.00
PAC	25.10	26.53	+1.43
PRA	0.71	0.74	+0.03
Tumor size	14.00	13.39	-0.61
18-OHF	1319.00	1404.11	+85.11
18-oxoF	88.90	109.26	+20.36

For reference, Table 2 collects the symbols used throughout the optimization sections so the appendix summary is not needed as a notation key. Tables generally use two-decimal displays, switching to scientific notation or additional digits where numerical tolerances and repeated-seed comparisons require them. Every numerical check and audit uses the stored machine-precision values in the companion `output/csv/` files; several artifacts also provide separate presentation-ready display columns.

3 Descriptive Statistics of the Predictors

Before any margin model is fitted, each predictor is summarized on its raw measurement scale. Table 3 reports completeness, central tendency, dispersion, range, and skewness for the ten numerical features across all 114 observations. No predictor contains a missing value in this synthetic cohort.

Table 2: Core notation.

Symbol	Meaning	Symbol	Meaning
$m = 114$	observations	$p = 10$	predictors
x_i	processed feature vector	$t_i \in \{-1, +1\}$	class sign
w, b	weights and intercept	ξ_i	soft-margin slack
α_i	dual multiplier	C	violation penalty
γ	RBF inverse squared bandwidth	K	kernel/Gram function

Table 3: Descriptive statistics for the ten numerical predictors on the raw measurement scale. IQR is the interquartile range and Range is the observed minimum to maximum. Skew is the adjusted Fisher–Pearson skewness; Skew (log) repeats it after the $\log(x + 10^{-6})$ transformation and is shown only for the four log-transformed hormone assays.

Predictor	Mean	Median	IQR	Range	Skew	Skew (log)
PAC	26.53	22.55	14.65–35.15	4.70–99.60	+1.45	-0.20
PRA	0.74	0.55	0.35–0.89	0.07–3.03	+1.75	-0.14
Potassium	3.59	3.60	3.23–3.90	2.60–4.70	+0.18	–
Tumor size	13.39	13.15	9.60–16.38	0.00–28.50	+0.22	–
18-OHF	1404.11	1105.50	692.75–1705.00	180.00–5994.00	+1.95	+0.14
18-oxoF	109.26	63.80	44.08–134.38	7.70–952.40	+3.81	+0.25
Systolic BP	139.42	141.00	130.00–149.75	100.00–193.00	+0.16	–
Diastolic BP	88.13	89.00	78.25–95.75	60.00–120.00	+0.13	–
DDD	2.31	2.00	2.00–3.00	0.00–5.00	-0.04	–
Age	49.36	49.00	43.00–56.75	21.00–72.00	-0.19	–

The skew (log) column is the empirical justification for the log transformation. The six non-log transformed predictors are close to symmetric, with skewness between -0.19 and 0.22 , whereas the four hormone assays are strongly right-skewed: 1.45 for PAC, 1.75 for PRA, 1.95 for 18-OHF, and 3.81 for 18-oxoF. After the log transformation all four fall within $[-0.20, 0.25]$. Only the four log-transformed hormones are transformed, and the decision follows from their marginal distributions alone, not from any outcome-dependent criterion.

Table 4 splits each predictor by subtype and reports two complementary quantities: the standardized mean difference (Cohen’s d) and the Welch t-statistic. They answer different questions. Cohen’s d measures *how large* the gap between the subtype means is, expressed in pooled standard deviations, and does not mechanically increase with cohort size. The Welch statistic measures *how precisely* that gap is resolved: it divides the same difference by its standard error, so, when the effect, variance, and class proportions remain fixed, its magnitude tends to grow in proportion to $\sqrt{m}$.

These contrasts are descriptive. Computed on the full cohort, they are not used to select features: every model in this report is fitted on all ten predictors inside the cross-validation pipeline. More importantly, a large univariate effect does not imply linear separability: d compares class means, whereas the hard-margin constraint of Section 4 requires *every individual observation* to lie on the correct side of one hyperplane. Figure 1 shows that even PAC, the strongest single predictor, leaves the two subtypes heavily overlapped.

Table 4: Predictor distributions by PA subtype. Cohen’s d is the standardized mean difference (UPA minus BPA) using the pooled standard deviation; Welch t is the same difference divided by its standard error, and p is the corresponding two-sided Welch-test p-value. These quantities are descriptive summaries of the synthetic cohort, not clinical inference.

Predictor	UPA mean (SD)	BPA mean (SD)	Cohen’s d	Welch t	p
PAC	34.9 (17.5)	18.4 (9.5)	+1.17	+6.20	< 0.001
PRA	0.5 (0.4)	1.0 (0.7)	-0.80	-4.32	< 0.001
Potassium	3.4 (0.4)	3.8 (0.5)	-0.95	-5.09	< 0.001
Tumor size	15.4 (4.8)	11.4 (5.9)	+0.74	+3.98	< 0.001
18-OHF	1883.0 (1225.7)	941.8 (557.0)	+0.99	+5.25	< 0.001
18-oxoF	148.1 (151.2)	71.8 (66.9)	+0.66	+3.46	< 0.001
Systolic BP	141.4 (16.3)	137.5 (17.5)	+0.23	+1.22	0.224
Diastolic BP	92.3 (10.5)	84.1 (11.3)	+0.75	+4.03	< 0.001
DDD	2.5 (0.9)	2.1 (1.0)	+0.36	+1.94	0.055
Age	48.6 (11.0)	50.1 (9.6)	-0.15	-0.80	0.425

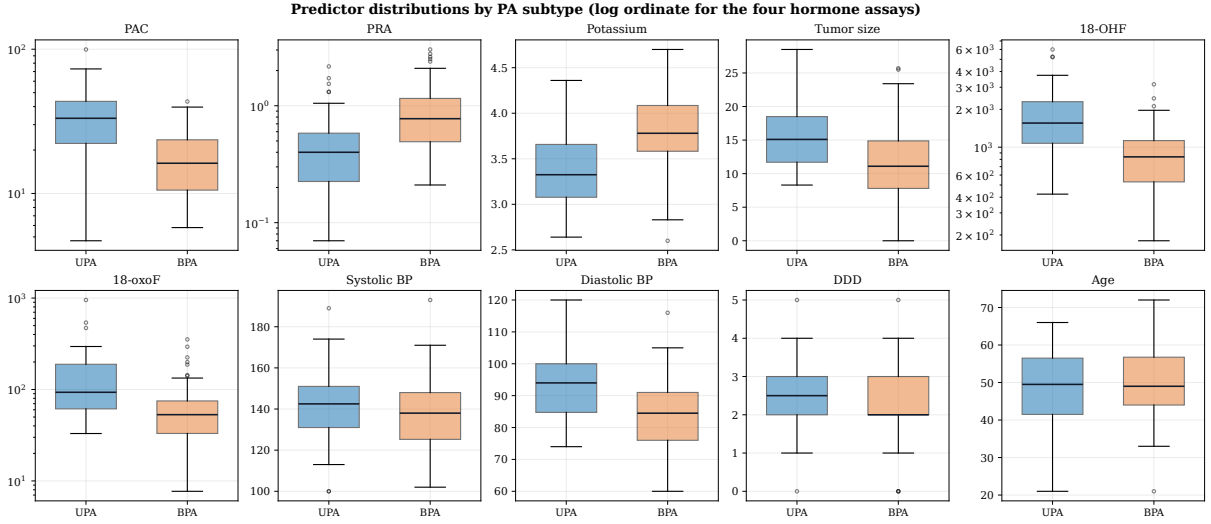


Figure 1: Distribution of each predictor by PA subtype. Boxes span the interquartile range with the median marked; whiskers extend to $1.5 \times \text{IQR}$ and points beyond are drawn individually. The four log transform features (PAC, PRA, 18-OHF, and 18-oxoF) use a logarithmic ordinate to make their right-skewed distributions legible. Every panel shows overlapping subtype distributions, which is why no single feature (and, as Section 4 shows, no linear combination of all ten) can separate the classes perfectly.

4 Hard-Margin SVM: Primal, Dual, and Existence

4.1 Primal Optimization Problem

The linear decision score is $s_i = w^\top x_i + b$, where $w \in \mathbb{R}^p$ is the normal vector of the separating hyperplane and $b \in \mathbb{R}$ is the intercept. The hard-margin primal SVM is the convex quadratic program [1, 6, 5]

$$\begin{aligned} \min_{w,b} \quad & \phi(w, b) = \frac{1}{2} \|w\|_2^2 \\ \text{subject to} \quad & t_i(w^\top x_i + b) \geq 1, \quad i = 1, \dots, m. \end{aligned} \quad (4.1)$$

The constraints require every training observation to be correctly classified with functional margin at least one. Minimizing $\|w\|_2^2/2$ maximizes the geometric margin $2/\|w\|_2$ between the two supporting hyperplanes [6].

In a hard-margin SVM there is no per-observation loss term to differentiate. Margin violations are forbidden by constraints rather than assigned a finite loss. Consequently, the quantity differentiated below is the *primal objective*, together with the affine constraint functions.

4.2 First and Second Derivatives

Define the augmented parameter vector, quadratic matrix, and signed augmented feature vector as

$$\theta = \begin{bmatrix} w \\ b \end{bmatrix}, \quad Q = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}, \quad u_i = t_i \begin{bmatrix} x_i \\ 1 \end{bmatrix}. \quad (4.2)$$

The objective and constraints can then be written as

$$\phi(\theta) = \frac{1}{2} \theta^\top Q \theta, \quad g_i(\theta) = 1 - u_i^\top \theta \leq 0. \quad (4.3)$$

The first derivative of the objective is

$$\nabla \phi(\theta) = Q\theta = \begin{bmatrix} w \\ 0 \end{bmatrix}, \quad (4.4)$$

and its second derivative is the constant Hessian

$$\nabla^2 \phi(\theta) = Q = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}. \quad (4.5)$$

For each margin constraint,

$$\nabla g_i(\theta) = -u_i = -t_i \begin{bmatrix} x_i \\ 1 \end{bmatrix}, \quad \nabla^2 g_i(\theta) = 0. \quad (4.6)$$

Because $Q \succeq 0$ and every g_i is affine, the primal problem is convex [5]. The objective is strictly convex in w but not in the complete vector (w, b) because the intercept is unpenalized. Thus the optimal w is unique whenever a solution exists, whereas the intercept need not be unique. There is also no hinge boundary or nondifferentiable point in this constrained primal formulation.

Section A.2 evaluates $u_i^\top \theta$ row by row in a ten-observation spreadsheet illustration.

4.3 Lagrangian and KKT Conditions

The derivatives of the primal objective describe only how $\|w\|_2^2/2$ changes; they do not by themselves enforce the margin constraints. Indeed, setting the objective gradient to zero would give $w = 0$, which is generally infeasible. The Lagrangian is therefore introduced as an auxiliary function, not as a replacement for the primal objective, to combine the objective with the constraints and express constrained optimality through the KKT conditions.

Introduce one multiplier $\alpha_i \geq 0$ for each inequality constraint. The Lagrangian is

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|_2^2 + \sum_{i=1}^m \alpha_i [1 - t_i(w^\top x_i + b)]. \quad (4.7)$$

Its first derivatives with respect to the primal variables are

$$\nabla_w \mathcal{L} = w - \sum_{i=1}^m \alpha_i t_i x_i, \quad \frac{\partial \mathcal{L}}{\partial b} = - \sum_{i=1}^m \alpha_i t_i. \quad (4.8)$$

At a stationary point these derivatives vanish, giving $w = \sum_i \alpha_i t_i x_i$ and $\sum_i \alpha_i t_i = 0$. Thus the separating direction is determined by the training observations whose constraints are active and whose multipliers are positive, rather than by the unconstrained minimum $w = 0$.

For fixed α , the Hessian with respect to (w, b) is again

$$\nabla_{(w,b)}^2 \mathcal{L} = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}. \quad (4.9)$$

When the primal is feasible, its Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for global optimality [2, 8]:

$$\begin{aligned} w - \sum_{i=1}^m \alpha_i t_i x_i &= 0, & \sum_{i=1}^m \alpha_i t_i &= 0, \\ t_i(w^\top x_i + b) - 1 &\geq 0, & \alpha_i &\geq 0, \\ \alpha_i [t_i(w^\top x_i + b) - 1] &= 0, & i &= 1, \dots, m. \end{aligned} \quad (4.10)$$

The final equality is complementary slackness. Read in both directions it states that an inactive constraint forces a zero multiplier ($t_i(w^\top x_i + b) - 1 > 0 \Rightarrow \alpha_i = 0$), while a positive multiplier forces an active constraint ($\alpha_i > 0 \Rightarrow t_i(w^\top x_i + b) = 1$); observations with $\alpha_i > 0$ therefore lie on a supporting margin and are the support vectors. Complementary slackness is also exactly the term that makes $\sum_i \alpha_i [t_i(w^\top x_i + b) - 1]$ vanish in the Lagrangian at optimality, which is what collapses the duality gap to $p^* = d^*$ [7]. Stationarity alone is not sufficient: primal feasibility, dual feasibility, and complementary slackness must also hold.

The KKT conditions are sufficient here because the objective is convex and the margin constraints are affine. They are also necessary whenever the hard-margin problem is feasible, for the reason established in Section 4.5. Conversely, if the data are not linearly separable, primal feasibility fails and no triplet (w, b, α) can satisfy the complete KKT system.

4.4 Dual Problem and the Meaning of the Dual Bound

The primal and dual optimize different variables and therefore point in opposite directions. The primal minimizes over the classifier parameters (w, b) . For any fixed multiplier vector $\alpha \geq 0$, define the dual function as the smallest Lagrangian value attainable by the primal variables,

$$q(\alpha) = \inf_{w,b} \mathcal{L}(w, b, \alpha). \quad (4.11)$$

For every primal-feasible (w, b) , the constraint terms $1 - t_i(w^\top x_i + b)$ are nonpositive. Hence

$$q(\alpha) \leq \mathcal{L}(w, b, \alpha) \leq \frac{1}{2} \|w\|_2^2. \quad (4.12)$$

Thus each dual-feasible α supplies a *lower bound* on every feasible primal objective value. This is the dual bound, the weak-duality inequality of [7]. The dual maximizes $q(\alpha)$ because it seeks the largest, and therefore tightest, lower bound; this explains why the primal is a minimization problem while the dual is a maximization problem.

Using stationarity, $w = \sum_i \alpha_i t_i x_i$ and $\sum_i \alpha_i t_i = 0$, eliminates w and b from the Lagrangian and gives

$$\begin{aligned} \max_{\alpha} \quad q(\alpha) &= \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j t_i t_j x_i^\top x_j \\ \text{subject to} \quad \alpha_i &\geq 0, \quad \sum_{i=1}^m \alpha_i t_i = 0. \end{aligned} \quad (4.13)$$

Weak duality always gives $d^* \leq p^*$. Whether that inequality tightens to an equality is a separate question, settled by the constraint qualification of the next subsection.

4.5 Slater's Condition and Strong Duality

Weak duality is automatic, but the dual is a faithful stand-in for the primal only when the gap closes [7]. For a convex program

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m, \\ & Ax = b, \end{aligned} \quad (4.14)$$

Slater's constraint qualification asks for a strictly feasible point: some $\tilde{x} \in \text{int } \mathcal{D}$ with

$$f_i(\tilde{x}) < 0 \quad \text{for all } i = 1, \dots, m, \quad A\tilde{x} = b, \quad (4.15)$$

where A has full row rank. When it holds, $p^* = d^*$; and if $d^* > -\infty$, the dual optimum is attained [2].

Two refinements decide the matter here. The strict inequality is needed only for the *non-affine* inequality constraints, so for a convex program whose inequality constraints are all affine, Slater's condition collapses to ordinary feasibility.

That is precisely the present case. The objective $\frac{1}{2} \|w\|_2^2$ is convex and every margin constraint $1 - t_i(w^\top x_i + b) \leq 0$ is affine in (w, b) , so strong duality holds for the hard-margin SVM as soon as one separating hyperplane exists. Feasibility is therefore the single hinge of the entire hard-margin analysis. The certificate of Section 5.1 shows the constraints are infeasible on this dataset, so $p^* = +\infty$ and there is no finite primal value for the dual to certify, which is why a dual solution must never be read as a trained classifier before feasibility is checked.

The soft-margin program of Section 6 differs in exactly the way that matters. Its constraints $1 - \xi_i - t_i(w^\top x_i + b) \leq 0$ and $-\xi_i \leq 0$ are again affine, and the point $(w, b, \xi) = (0, 0, \mathbf{1})$ is feasible for *any* dataset. Slater's condition therefore holds unconditionally, strong duality is guaranteed in advance, and the vanishing primal–dual gaps reported later (4.9×10^{-10} for the linear model, 2.2×10^{-11} for the RBF model) are genuine checks on solver accuracy rather than assumptions being taken on trust.

4.6 Linear Separability and Existence of a Hard-Margin Solution

Let $U \in \mathbb{R}^{m \times (p+1)}$ have rows $u_i^\top = t_i[x_i^\top, 1]$ and let $\theta = [w^\top, b]^\top$. A hard-margin solution exists if and only if the linear system

$$U\theta \geq \mathbf{1} \quad (4.16)$$

is feasible. This condition is also equivalent to strict linear separability: if some hyperplane correctly separates all observations with strictly positive signed scores, rescaling (w, b) makes the smallest signed score equal to one; conversely, any point satisfying the unit-margin inequalities is a strict separator.

The existence question is checked by the zero-objective linear program

$$\min_{\theta} 0 \quad \text{subject to} \quad -U\theta \leq -\mathbf{1}. \quad (4.17)$$

If it is feasible, its output supplies a valid starting point for minimizing the quadratic primal objective. If it is infeasible, changing the objective from zero to $\|w\|_2^2/2$ cannot make the empty feasible set nonempty: the hard-margin primal has no solution. This logical order separates two questions that are often confused: feasibility asks whether any separating hyperplane exists, whereas optimization selects the maximum-margin hyperplane only after existence has been established.

Two worked numerical examples are provided in Appendix A. They illustrate feasible hard-margin primal–dual calculations without interrupting the full-cohort feasibility argument.

5 Hard-Margin Feasibility Audit and Certificate

5.1 Full-Data Infeasibility Certificate

Three mutually reinforcing checks are used on the complete processed dataset, in the order direct solver status, algebraic Farkas certificate, and geometric convex-hull interpretation. Here $U \in \mathbb{R}^{114 \times 11}$ contains the ten standardized features and the intercept column.

Direct full-data feasibility check. The primal model supplied to a constrained solver is

$$\begin{aligned} \min_{w, b} \quad & \frac{1}{2} \|w\|_2^2 \\ \text{subject to} \quad & U\theta \geq \mathbf{1}. \end{aligned} \quad (5.1)$$

Before an objective can be minimized, the solver must find a point satisfying all 114 constraints. The implementation therefore first solves the equivalent feasibility model $\min_{\theta} 0$ subject to $-U\theta \leq -\mathbf{1}$; changing the objective cannot make an empty feasible set nonempty. The direct solver audit returned the result in Table 5.

Thus the correct statement is that the solver returns an *infeasible status*; it does not return an “infeasible solution.” No primal solution exists for the solver to report. This is the direct computational verification used in the direct verification workflow. In the corresponding hard-margin dual, primal infeasibility means that no finite primal optimum exists to be matched by a valid classifier. Indeed, scaling the Farkas vector below gives a dual-feasible ray with zero quadratic term and an arbitrarily large linear term, so the hard-margin dual is unbounded above; it cannot rescue the empty primal feasible set.

Table 5: Direct full-data primal feasibility check.

Item	Solver result
Decision variables $(w_1, \dots, w_{10}, b)$	11
Hard-margin constraints	114
Solver	SciPy HiGHS
HiGHS model status	Status 8: Infeasible
Primal parameters (w, b) returned	None
Primal objective $\frac{1}{2}\ w\ _2^2$	Not defined

Independent certificate. The solver status is then strengthened by a direct, independently checkable proof. Farkas’ lemma states that the system $U\theta \geq \mathbf{1}$ is infeasible if and only if there is a vector $\lambda \in \mathbb{R}^{114}$ satisfying

$$\lambda \geq 0, \quad U^\top \lambda = 0, \quad \mathbf{1}^\top \lambda = 1. \quad (5.2)$$

The auxiliary linear program for this certificate returned twelve nonzero weights, shown in Table 6; the other 102 weights are zero. The displayed values alone should not be used to recompute the certificate; every check uses the unrounded values in the companion CSV file.

Table 6: Nonzero weights in the full-data Farkas infeasibility certificate.

Patient	Class	λ_i	Patient	Class	λ_i
CGH-021	UPA	0.21	CGH-072	BPA	0.04
CGH-024	BPA	0.05	CGH-081	UPA	0.01
CGH-026	BPA	0.16	CGH-092	BPA	0.01
CGH-053	UPA	0.04	CGH-105	UPA	0.01
CGH-058	UPA	0.17	CGH-108	BPA	0.05
CGH-060	BPA	0.18	CGH-111	UPA	0.07

The full-precision numerical audit gives

$$\begin{aligned} \mathbf{1}^\top \lambda &= 1.00, \\ \sum_{i:t_i=+1} \lambda_i &= 0.50, \\ \sum_{i:t_i=-1} \lambda_i &= 0.50, \\ \|U^\top \lambda\|_\infty &< 10^{-15}. \end{aligned} \quad (5.3)$$

The final residual is reported as a tolerance statement because its exact machine-zero mantissa can change with the numerical backend. Now suppose, for contradiction, that a hard-margin parameter θ satisfied $U\theta \geq \mathbf{1}$. Nonnegativity of λ would imply

$$\underbrace{(U^\top \lambda)^\top}_0 \theta = \lambda^\top U\theta \geq \lambda^\top \mathbf{1} = 1, \quad (5.4)$$

which is the contradiction $0 \geq 1$. Therefore the complete hard-margin constraint system has no feasible solution. In fact, the twelve observations in Table 6 already provide a sufficient infeasible subsystem.

The same certificate has a geometric interpretation. Because the intercept component of $U^\top \lambda = 0$ gives equal class masses of 0.5, define $\beta_i^+ = 2\lambda_i$ for UPA and $\beta_i^- = 2\lambda_i$ for BPA. Each set of coefficients is nonnegative and sums to one, while the feature components of $U^\top \lambda = 0$ give

$$\sum_{i:t_i=+1} \beta_i^+ x_i = \sum_{i:t_i=-1} \beta_i^- x_i. \quad (5.5)$$

Thus the standardized UPA and BPA convex hulls contain the same point. Their largest coordinate difference is below 10^{-15} (machine precision), as illustrated in Figure 2. Intersecting convex hulls cannot be strictly separated by a hyperplane.

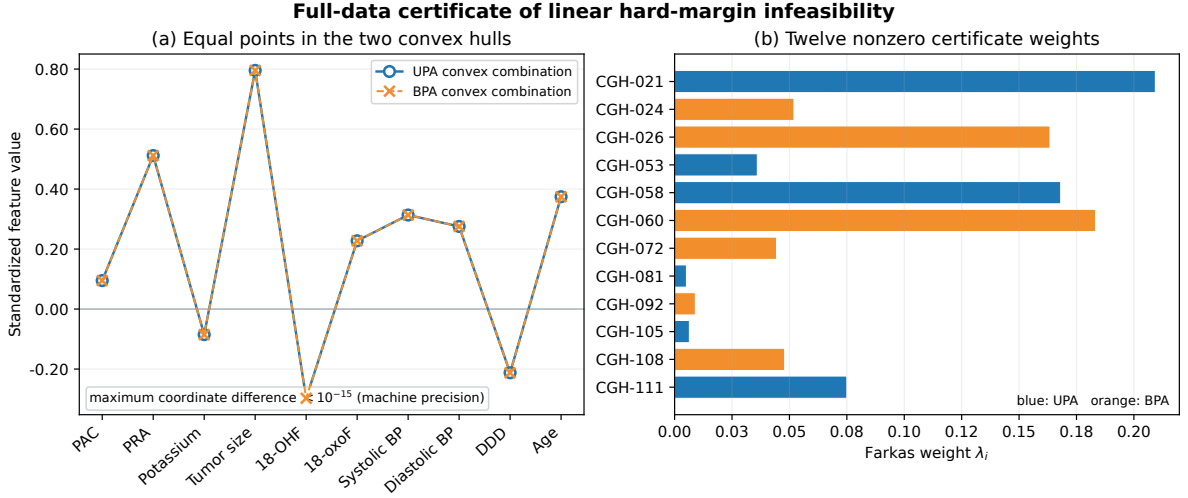


Figure 2: Full-data infeasibility certificate. Panel (a) shows the identical ten-dimensional point obtained as a convex combination of UPA observations and as a convex combination of BPA observations; the plot compares coordinates directly and is not a two-dimensional projection. Panel (b) shows the twelve nonzero Farkas weights.

Two complementary illustrations make the certificate intuitive. Figure 3 contrasts the two systems of Farkas' lemma: either a separating hyperplane exists (System A), or a nonnegative combination of the constraints proves a contradiction $0 \geq 1$ (System B), never both. Figure 4 provides a two-dimensional visual intuition by projecting the project data onto its first two principal components. The projected convex hulls overlap, which is consistent with—but does not by itself prove—nonseparability in the original ten-dimensional space. The proof is the full-dimensional Farkas certificate above. The theory-of-the-alternative structure behind this argument follows the treatment of duality and first-order optimality in the course lectures [7, 8].

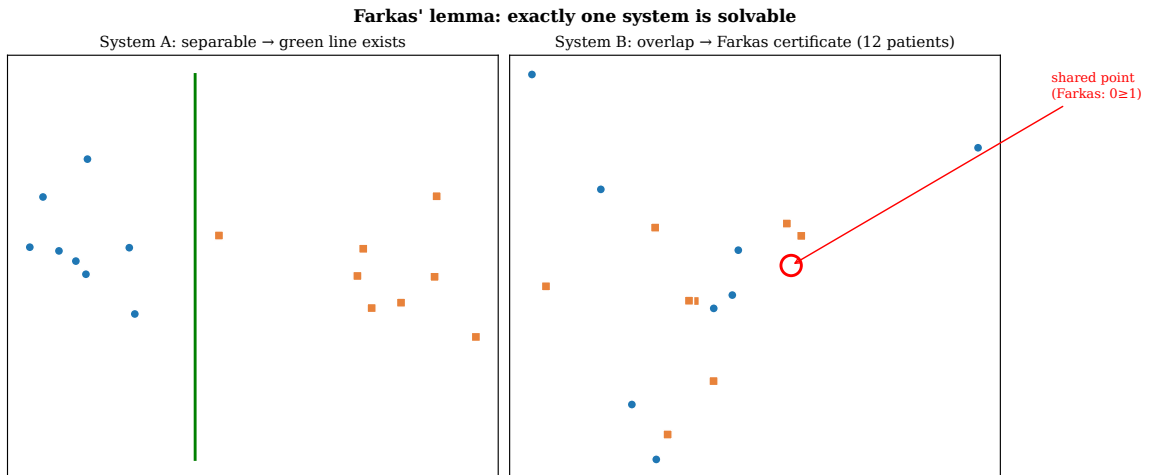


Figure 3: Farkas' lemma as a theorem of the alternative. System A (left): the classes are separable, so a hyperplane exists. System B (right): the classes overlap, so a nonnegative weighting of the points certifies infeasibility ($0 \geq 1$). Exactly one system is solvable; for the project data it is System B.

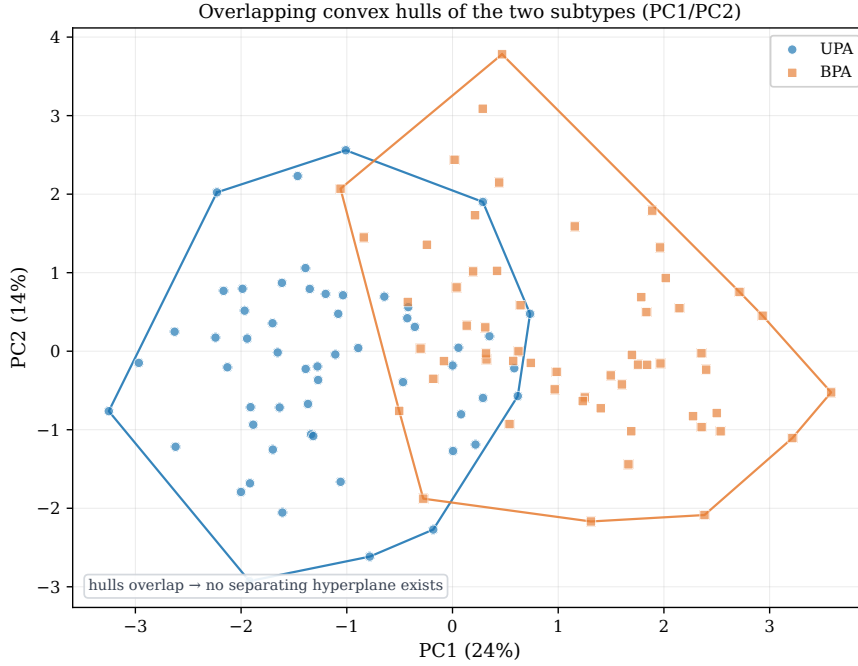


Figure 4: Convex hulls of the UPA and BPA subtypes in the first two principal components of the standardized data. Their overlap illustrates the class mixing visible in this projection but does not establish nonseparability in ten dimensions. The full-dimensional certificate in Figure 2 provides that proof.

Consequently, for the current preprocessed project data,

$$\boxed{\text{the UPA and BPA classes are not linearly separable in this processed representation.}} \quad (5.6)$$

The qualification matters: the log transform is nonlinear, so this certificate does not make a claim about every alternative representation of the raw data. Nor does linear nonseparability preclude separation after a nonlinear kernel map; it establishes only that the stated linear hard-margin problem is empty. The certificate, CSV output, numerical checks, and figure are reproduced by `scripts/build_full_data_infeasibility_certificate.py`. The script writes four machine-readable audit files under `output/csv/`: the direct solver status, sparse Farkas vector, two convex-hull representations, and numerical residual checks. The two tables above are generated from the same Python results and then included in this PDF.

5.2 Cross-Validation Feasibility Results and ROC Rule

The feasibility audit uses stratified five-fold cross-validation with shuffling and random seed zero. All preprocessing operations are fitted inside the training fold. In each fold, the hard-margin classifier is fitted only if the training constraints are feasible. Held-out decision scores would then be pooled to construct the ROC curve, and AUROC would summarize discrimination across score thresholds.

The result is unambiguous: the full-data problem is infeasible, and all five training-fold problems are also infeasible. Thus linear separation is not available in any evaluated hard-margin fit; there are no valid hard-margin parameters with which to score the corresponding held-out observations.

This ordering is important. ROC analysis requires a fitted scoring function for every held-out observation. If a training fold is infeasible, there are no valid hard-margin parameters for that

Table 7: Hard-margin feasibility results for the full synthetic cohort and the five cross-validation training folds. “No” means that no parameter pair (w, b) satisfies every unit-margin constraint in that row’s training set.

Data split	Training observations	Held-out observations	Feasible
Full dataset	114	–	No
Fold 1	91	23	No
Fold 2	91	23	No
Fold 3	91	23	No
Fold 4	91	23	No
Fold 5	92	22	No

fold and therefore no legitimate out-of-fold scores. Reporting an ROC curve in that situation would require silently changing either the model or the data.

The worked subsystem in Section A.2 is not a cross-validation split and is not used to generate ROC results.

The hard-margin question is therefore closed: the complete cohort and every training fold are infeasible, so the model cannot be fitted or evaluated by ROC analysis. Section 6 now relaxes the unit-margin constraints explicitly by introducing penalized slack variables.

6 Linear Soft-Margin SVM

6.1 Why Slack Variables Are Needed

The preceding hard-margin audit, including the certificate in Section 5.1, establishes that no (w, b) can satisfy all 114 unit-margin inequalities simultaneously. The soft-margin model changes the assumption rather than attempting to optimize over an empty set. For each observation it introduces a slack variable $\xi_i \geq 0$ that measures the shortfall from the desired signed functional margin of one:

$$\xi_i = \max\{0, 1 - t_i(w^\top x_i + b)\}. \quad (6.1)$$

When $\xi_i = 0$, the observation lies on or outside its supporting margin. If $0 < \xi_i \leq 1$, it lies inside the margin but remains on the correct side of the decision boundary. If $\xi_i > 1$, its signed score is negative and it is misclassified. This interpretation directly connects the optimization variables to the geometry of overlapping UPA and BPA observations.

6.2 Soft-Margin Primal Problem

The linear soft-margin primal problem is [1, 6]

$$\begin{aligned} \min_{w, b, \xi} \quad & p(w, b, \xi) = \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^m \xi_i \\ \text{subject to} \quad & t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad i = 1, \dots, m, \\ & \xi_i \geq 0, \quad i = 1, \dots, m. \end{aligned} \quad (6.2)$$

The first term favors a wide margin, while the second penalizes total margin violation. The constant $C > 0$ controls their trade-off. Unlike the hard-margin problem, this primal is always feasible: $w = 0$, $b = 0$, and $\xi_i = 1$ satisfy every constraint. Eliminating the slack variables gives

the equivalent unconstrained hinge-loss form

$$\min_{w,b} \quad \frac{1}{2}\|w\|_2^2 + C \sum_{i=1}^m \max\{0, 1 - t_i(w^\top x_i + b)\}. \quad (6.3)$$

The constrained form is retained below because it makes the primal–dual relationship and the role of C explicit.

6.3 Lagrangian and KKT Conditions

Introduce $\alpha_i \geq 0$ for $1 - \xi_i - t_i(w^\top x_i + b) \leq 0$ and $\mu_i \geq 0$ for $-\xi_i \leq 0$. The Lagrangian is

$$\begin{aligned} \mathcal{L}(w, b, \xi, \alpha, \mu) = & \frac{1}{2}\|w\|_2^2 + C \sum_i \xi_i \\ & + \sum_i \alpha_i [1 - \xi_i - t_i(w^\top x_i + b)] - \sum_i \mu_i \xi_i. \end{aligned} \quad (6.4)$$

Stationarity with respect to the primal variables gives

$$w = \sum_i \alpha_i t_i x_i, \quad \sum_i \alpha_i t_i = 0, \quad C - \alpha_i - \mu_i = 0. \quad (6.5)$$

Because $\mu_i \geq 0$, the final equality produces the upper bound $0 \leq \alpha_i \leq C$. The complete KKT system additionally requires primal feasibility, dual feasibility, and complementary slackness:

$$\begin{aligned} t_i(w^\top x_i + b) - 1 + \xi_i &\geq 0, & \xi_i &\geq 0, \\ 0 &\leq \alpha_i \leq C, \\ \alpha_i [1 - \xi_i - t_i(w^\top x_i + b)] &= 0, \\ (C - \alpha_i) \xi_i &= 0. \end{aligned} \quad (6.6)$$

These conditions are necessary and sufficient because the primal objective is convex, its constraints are affine, and strict feasibility is obtained by choosing sufficiently large positive slacks. They also explain the support vectors. An observation with $\alpha_i = 0$ does not determine w and lies outside the margin. A point with $0 < \alpha_i < C$ lies exactly on a supporting margin. A point with $\alpha_i = C$ may lie inside the margin or be misclassified, so it receives the largest multiplier permitted by the chosen penalty.

6.4 Soft-Margin Dual Problem

Minimizing the Lagrangian over w , b , and ξ using the stationarity conditions yields

$$\begin{aligned} \max_{\alpha} \quad d(\alpha) = & \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j t_i t_j x_i^\top x_j \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C, \quad \sum_{i=1}^m \alpha_i t_i = 0. \end{aligned} \quad (6.7)$$

The objective has the same form as the hard-margin dual, but the new box constraint $\alpha_i \leq C$ is essential. It prevents any single observation from exerting unlimited pressure on the separating hyperplane. Since the soft-margin primal is strictly feasible, strong duality holds and the optimal primal and dual values are equal up to numerical solver tolerance.

6.5 Selecting the Penalty Parameter by Grid Search

A small C places greater emphasis on a wide, strongly regularized margin and accepts more slack. A large C penalizes violations more severely and moves toward hard-margin behavior, which can increase sensitivity to overlapping observations. Because neither extreme is known in advance, C is selected from the half-logarithmic grid

$$\mathcal{C} = \{10^{k/2} : k = -4, -3, \dots, 4\}. \quad (6.8)$$

Displayed to two decimal places, these values are

$$0.01, 0.03, 0.10, 0.32, 1.00, 3.16, 10.00, 31.62, 100.00. \quad (6.9)$$

Each candidate is evaluated by mean five-fold stratified validation AUROC. The log transformation, median imputation, standardization, and linear SVM are placed in one pipeline, so every preprocessing parameter is estimated from the corresponding training fold only.

There are therefore two distinct optimization levels. For a fixed candidate C and training fold r , the SMO solver first finds the optimal model for that value alone:

$$(w_{C,r}^*, b_{C,r}^*, \xi_{C,r}^*) = \arg \min_{w,b,\xi} \left\{ \frac{1}{2} \|w\|^2 + C \sum_{i \in \mathcal{T}_r} \xi_i \right\} \quad \text{subject to the fold's soft-margin constraints.} \quad (6.10)$$

That optimized model scores validation fold $\mathcal{V}_r$. Grid search then selects the hyperparameter by

$$\hat{C} = \arg \max_{C \in \mathcal{C}} \frac{1}{5} \sum_{r=1}^5 \text{AUROC}(\mathcal{V}_r; w_{C,r}^*, b_{C,r}^*). \quad (6.11)$$

Thus every candidate C has its own fully optimized coefficients in every fold; C is compared using held-out AUROC, not by comparing training objective values across different penalties. After $\hat{C}$ is chosen, the pipeline solves once more at that fixed value using all observations available to that stage.

Nested cross-validation separates model selection from performance estimation. In each of five outer folds, an independent five-fold inner grid search chooses C using only the outer training observations. The selected model then scores the untouched outer test fold. Table 8 shows that all five inner searches selected $C = 0.01$. Because the same seed-zero outer partition is also used to draw the fixed- C validation curve, and every inner search selected the same C , its five test fold AUROCs numerically coincide with the corresponding fixed- C fold scores. The nesting remains methodologically valid—each outer test fold is untouched by selection—but the numerical coincidence is not independent corroborating evidence.

Table 8: Nested cross-validation for soft-margin penalty selection. AUROC values and C are displayed to two decimal places.

Outer fold	Train	Test	Selected C	Inner AUROC	Outer AUROC
1	91	23	0.01	0.95	0.92
2	91	23	0.01	0.96	0.95
3	91	23	0.01	0.94	0.98
4	91	23	0.01	0.91	0.98
5	92	22	0.01	0.95	0.89
Mean	—	—	—	0.941	0.945

6.6 Soft-Margin Solution on the Project Dataset

The final full-data grid search also selected $C = 0.01$, with mean five-fold validation AUROC 0.95. Figure 5 shows both the validation curve over C and the nested out-of-fold ROC curve. Across the five untouched outer folds in this illustrative seed-zero partition, mean fold-wise AUROC is 0.95; pooling all out-of-fold scores gives AUROC 0.95, with accuracy, sensitivity, and specificity all approximately 0.86 at decision threshold zero. These numbers are generated by the explicit linear soft-margin primal–dual model defined in this section.

The fitted standardized-space coefficients in Table 9 give the final linear scoring function. Their signs describe the direction of the score after the stated preprocessing but should not be interpreted as causal clinical effects.

Table 9: Final standardized-space soft-margin coefficients at $C = 0.01$.

Parameter	Estimate	Parameter	Estimate
PAC	0.29	18-oxoF	0.22
PRA	-0.18	Systolic BP	0.09
Potassium	-0.19	Diastolic BP	0.18
Tumor size	0.15	DDD	0.07
18-OHF	0.23	Age	-0.08
Intercept	-0.02		

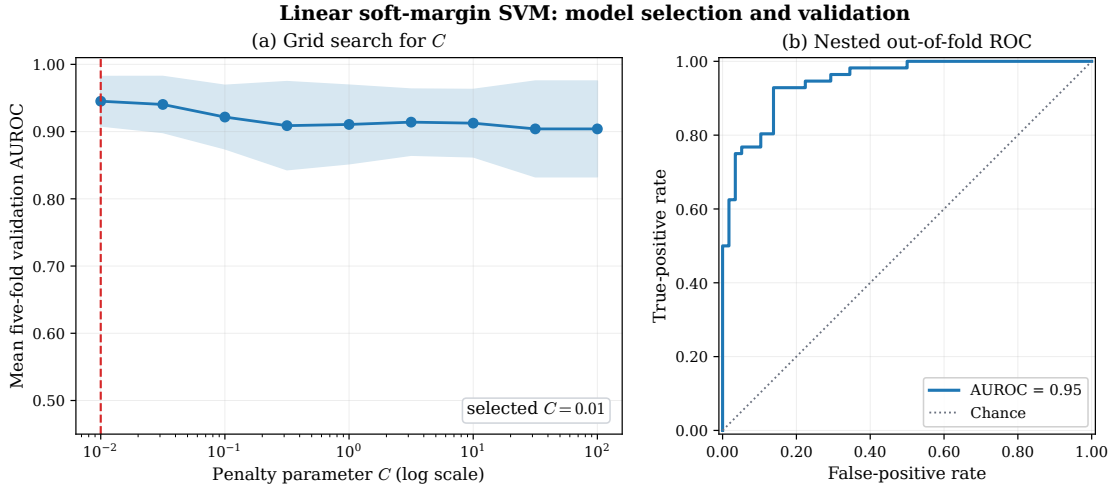


Figure 5: Soft-margin model selection and validation. Panel (a) reports mean five-fold validation AUROC with one population standard deviation across folds ($\text{ddof} = 0$) for every candidate C ; the dashed line marks the selected value. Panel (b) reports the nested out-of-fold ROC obtained when C is selected inside each outer training fold.

The full-data solution contains 82 support vectors and 77 observations with positive slack, with $\sum_i \xi_i = 46.89$. Substitution into the primal gives

$$\begin{aligned}
 p^* &= \frac{1}{2} \|w\|_2^2 + C \sum_i \xi_i \\
 &= \frac{1}{2} (0.33) + (0.01)(46.89) \\
 &\approx 0.63.
 \end{aligned} \tag{6.12}$$

The dual calculation gives $d^* \approx 0.63$, and the full-precision primal–dual gap is 4.85×10^{-10} . The weight-reconstruction residual is zero at stored precision. The dual equality residual is below 10^{-15} and can vary slightly with floating-point summation order. The two maximum complementary-slackness residuals are 2.35×10^{-10} and 8.57×10^{-11} , confirming the KKT conditions to solver tolerance. Table 10 summarizes the model-selection and optimization results.

Table 10: Soft-margin model-selection, validation, and optimization summary for the prespecified seed-zero split.

Quantity	Value
Selected C from full-data grid search	0.01
Mean inner-CV AUROC at selected C	0.95
Nested out-of-fold AUROC	0.95
Nested out-of-fold accuracy	0.86
Nested sensitivity	0.86
Nested specificity	0.86
Full-data support vectors	82
Full-data observations with $\xi_i > 0$	77
Full-data $\sum_i \xi_i$	46.89
Full-data primal objective	0.63
Full-data dual objective	0.63

All full-precision grid-search results, nested predictions, coefficients, support-vector multipliers, slacks, and primal–dual checks are written under `output/csv/` by `scripts/linear_soft_margin_svm.py`. Tables in this section and both panels of Figure 5 are generated from those Python outputs.

The penalty C is selected by the grid search of Figure 6: validation AUROC is highest at the strongly regularized end and the training–validation gap is smallest there, so $C = 0.01$ is preferred. Figure 7 recasts the same grid as a Pareto trade-off between validation performance and overfitting, on which $C = 0.01$ is the single non-dominated point within the prespecified grid. These AUROC differences are small relative to the fold-to-fold standard deviations and should not be read as precise evidence that adjacent settings differ. Figure 8 shows the resulting soft-margin boundary in the first two principal components: the separating line passes straight through the region where the two subtypes overlap, accepting the misclassifications it cannot avoid and charging for them through the slack variables.

6.7 Solving the Primal Directly: Stochastic Subgradient and Momentum Methods

The dual quadratic program above is solved through `scikit-learn`’s LIBSVM-based SVC implementation [16, 17]; LIBSVM uses an SMO-type decomposition strategy related to Platt’s algorithm [15]. The soft-margin *primal* can instead be minimized directly. Eliminating the slack variables gives hinge losses $\ell_i(w, b) = \max(0, 1 - t_i(w^\top x_i + b))$. Dividing the constrained objective by the positive constant mC does not change its minimizer and gives

$$P(w, b) = \frac{1}{m} \sum_{i=1}^m \ell_i(w, b) + \frac{\lambda}{2} \|w\|^2, \quad \lambda = \frac{1}{mC}. \quad (6.13)$$

This regularized finite sum is a natural target for the stochastic *subgradient* method (SSG), often grouped under the broader label SGD; Pegasos is a standard SVM example of this approach

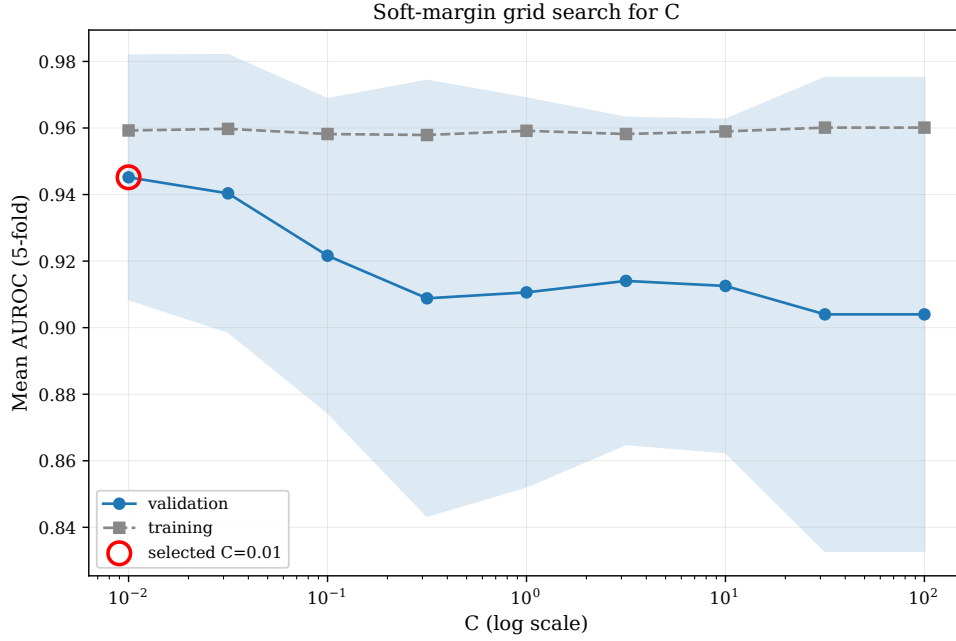


Figure 6: Grid search for the penalty C . Mean five-fold validation AUROC (with training AUROC for reference) across the C grid; the selected value $C = 0.01$ is circled. Stronger regularization generalizes best on this overlapping data.

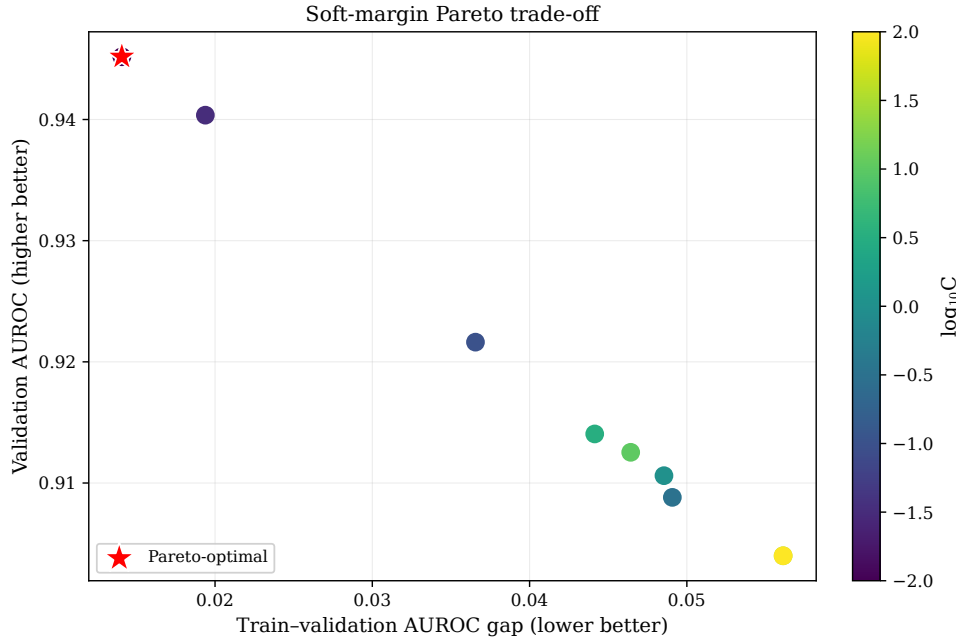


Figure 7: The same grid as a Pareto trade-off between mean validation AUROC (higher is better) and the train–validation AUROC gap (lower is better). Points are colored by $\log_{10} C$; $C = 0.01$ (star) is the single Pareto-optimal setting.

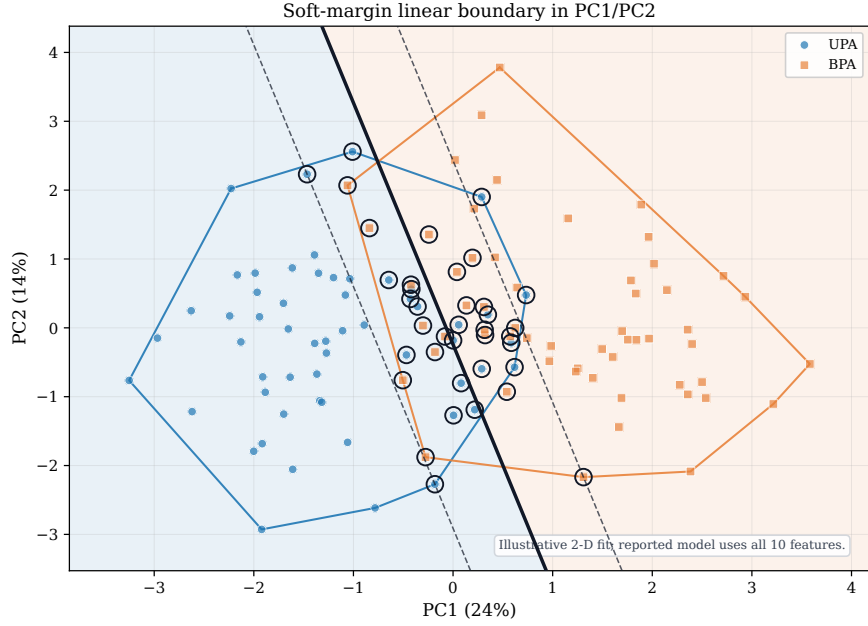


Figure 8: Fitted soft-margin decision boundary in the first two principal components, with the ± 1 margins and the circled support vectors. The boundary is an illustrative two-dimensional fit; the reported model is trained on all ten features.

[12]. The distinction matters because hinge loss is not differentiable when $t_i(w^\top x_i + b) = 1$. For implementation, let $\theta = [w^\top, b]^\top$, $z_i = [x_i^\top, 1]^\top$, and $R = \text{diag}(1, \dots, 1, 0)$. The zero in R ensures that the intercept is not regularized. Each step samples one index k uniformly at random. A valid stochastic subgradient is

$$g_k(\theta) = \begin{cases} \lambda R \theta, & t_k \theta^\top z_k \geq 1, \\ \lambda R \theta - t_k z_k, & t_k \theta^\top z_k < 1. \end{cases} \quad (6.14)$$

The regularizer is λ -strongly convex in the weight coordinates [9], while the intercept remains unpenalized. Consequently, a global strong-convexity guarantee for the full (w, b) vector does not apply verbatim; the diminishing schedule below is a practical heuristic for this formulation. The baseline update is the stochastic subgradient method from the course lecture [10], $\theta_{k+1} = \theta_k - \eta_k g_k(\theta_k)$. Two momentum variants from the momentum-methods lecture [11] are also evaluated; their classical origins are Polyak’s Heavy Ball method and Nesterov’s accelerated method [13, 14]:

$$\begin{aligned} \text{Heavy Ball: } \theta_{k+1} &= \theta_k - \eta_k g_k(\theta_k) + \beta(\theta_k - \theta_{k-1}), \\ \text{Nesterov-style: } y_k &= \theta_k + \beta(\theta_k - \theta_{k-1}), \\ \theta_{k+1} &= y_k - \eta_k g_k(y_k). \end{aligned} \quad (6.15)$$

The word “style” is important: classical Heavy Ball and Nesterov acceleration theory assumes a smooth gradient, whereas the hinge objective is nonsmooth. Replacing the gradient by a sampled subgradient gives useful empirical comparators, but does not inherit the classical accelerated convergence rate.

The optimizer comparison begins only after model selection. The full-data grid search above selected

$$\boxed{C = 0.01}, \quad \lambda = \frac{1}{mC} = \frac{1}{114(0.01)} \approx 0.8772. \quad (6.16)$$

Figure 9 and Table 11 therefore optimize the primal for *this one fixed value*, $C = 0.01$. They do not repeat the grid search and do not compare different values of C . The reference optimum

for the normalized objective is $P^* = 0.554274$, obtained by solving the equivalent slack-variable convex quadratic program at $C = 0.01$. Multiplying back by $mC = 1.14$ gives the original primal value 0.6319, agreeing with the primal–dual solution reported above. For a controlled comparison, all three algorithms start at zero, use the same 50,000 sampled observations, and use $\eta_k = a/[\lambda(k + 100)]$. A small common grid $a \in \{0.25, 0.5, 1, 2\}$ is searched for every method; the momentum methods also search $\beta \in \{0.3, 0.5, 0.7, 0.9\}$. Within each method, the pair with the lowest final averaged-iterate objective on a common tuning sequence is reported. The selected settings are then rerun on ten new, common sampling sequences to assess optimization robustness. This tuning is only an optimization diagnostic on the training objective and is not a predictive performance estimate.

Table 11: Controlled optimizer comparison for the single grid-selected value $C = 0.01$ ($\lambda \approx 0.8772$). The 5% and 1% columns give the first step at which the best raw iterate reaches the stated gap relative to P^* ; 114 steps correspond to approximately one epoch. AUROC is in-sample and is shown only as a solution-agreement diagnostic. Cosine is the similarity of the averaged iterate to the reference QP solution.

Method	a	β	$P(\bar{w}_K)$	5% steps	1% steps	AUROC	Cosine
SSG	2.00	—	0.5544	96	389	0.959	0.9998
Heavy Ball	0.25	0.9	0.5544	150	391	0.959	0.9998
Nesterov-style	2.00	0.3	0.5544	98	389	0.959	0.9998

Because a single sampled-observation sequence can make a stochastic optimizer look unusually favorable, Table 12 repeats the already selected optimizer settings on ten new common sampling sequences. Medians and interquartile ranges (IQRs) are descriptive optimization diagnostics.

Table 12: Robustness over ten new common sampling sequences at the single fixed value $C = 0.01$. Hyperparameters were selected on the separate seed-2026 sequence.

Method	Final gap, median [IQR]	5% steps, median [IQR]	1% steps, median [IQR]
SSG	1.45e-04 [1.30e-04, 1.82e-04]	41 [34, 48]	260 [186, 353]
Heavy Ball	1.49e-04 [1.42e-04, 1.91e-04]	60 [35, 73]	519 [296, 558]
Nesterov-style	1.54e-04 [1.44e-04, 1.99e-04]	39 [29, 50]	356 [256, 510]

The illustrative seed-2026 comparison does not show a meaningful momentum advantage. SSG reaches the 5% gap in 96 updates (0.84 epochs) and the 1% gap in 389 updates (3.41 epochs). Heavy Ball reaches the two thresholds in 150 and 391 updates, while the Nesterov-style method requires 98 and 389 updates. All averaged iterates have objective approximately 0.5544, in-sample AUROC approximately 0.959, and cosine similarity above 0.9998 to the reference solution. Thus the methods converge to essentially the same fixed- C classifier, while plain SSG is the simplest and at least as fast under this controlled run. This negative acceleration result is plausible: strong convexity does not remove the hinge kink, sampled subgradients remain noisy near the optimum, and the smooth-gradient Heavy Ball and Nesterov rate guarantees do not apply. Running averages stabilize all three methods. Across the ten new sequences, median final gaps are 1.45×10^{-4} for SSG, 1.49×10^{-4} for Heavy Ball, and 1.54×10^{-4} for Nesterov-style. Median steps to the 1% threshold are 260, 519, and 356, respectively. The sequence-to-sequence variation reinforces the conclusion: none of the momentum variants provides a robust acceleration advantage under this fixed budget and tuning protocol.

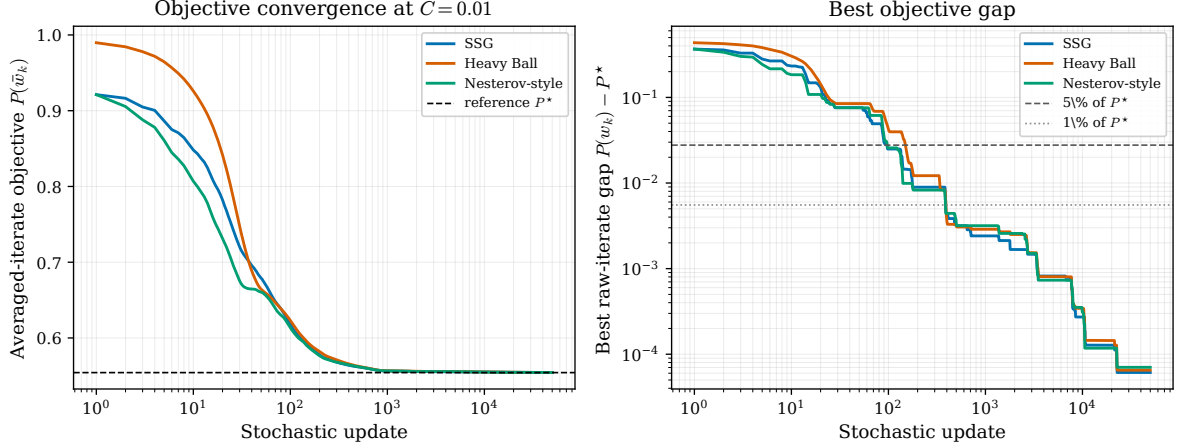


Figure 9: Stochastic primal optimization for the fixed selected value $C = 0.01$ ($\lambda \approx 0.8772$), under a shared data sequence and iteration budget. Left: objective values of the running averaged iterates. Right: best-so-far raw-iterate optimality gaps. All three methods approach the same reference optimum, but momentum does not accelerate this nonsmooth stochastic problem under the tested settings.

7 Kernel Trick: RBF Soft-Margin SVM

7.1 From a Linear Boundary to a Kernel Boundary

The strong linear soft-margin result does not prove that a nonlinear boundary is unnecessary; it only establishes a competitive baseline. The kernel trick tests nonlinear structure without explicitly constructing a potentially high-dimensional feature vector. Let $\varphi(x)$ map an observation into a feature space $\mathcal{H}$. The soft-margin primal in that space is

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|_{\mathcal{H}}^2 + C \sum_{i=1}^m \xi_i \\ \text{subject to} \quad & t_i(\langle w, \varphi(x_i) \rangle_{\mathcal{H}} + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \end{aligned} \quad (7.1)$$

The corresponding dual depends on transformed observations only through inner products:

$$\begin{aligned} \max_{\alpha} \quad & \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j t_i t_j K(x_i, x_j) \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i t_i = 0, \end{aligned} \quad (7.2)$$

where $K(x_i, x_j) = \langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}}$. The fitted score is

$$f(x) = \sum_{i \in \mathcal{S}} \alpha_i t_i K(x_i, x) + b, \quad (7.3)$$

so only the support-vector set $\mathcal{S}$ is needed at prediction time. Kernelization does not replace the soft margin: C still bounds each dual multiplier and controls the penalty for overlap.

This analysis uses the radial basis function (RBF) kernel

$$K(x_i, x_j) = \exp[-\gamma \|x_i - x_j\|_2^2]. \quad (7.4)$$

Small γ produces a smooth, slowly varying decision surface and tends toward linear-like behavior over a limited data region. Large γ makes each observation influential only in a narrow neighborhood, allowing a highly curved boundary that can memorize a small training cohort. Therefore C and γ must be tuned jointly.

7.2 Validity of the RBF Kernel: Positive Semi-Definiteness and Feature Map

Everything above presumes that some feature map φ exists with $K(x, y) = \langle \varphi(x), \varphi(y) \rangle_{\mathcal{H}}$. That is not automatic: an arbitrary similarity function need not be an inner product in any space, and if it is not, the dual loses concavity and the optimization guarantees of Section 4.5 no longer apply. This subsection establishes the two properties that license the RBF kernel.

A symmetric kernel K is *positive semi-definite* (PSD) if for every finite set $x_1, \dots, x_m$ the Gram matrix $G_{ij} = K(x_i, x_j)$ satisfies $c^\top G c \geq 0$ for all $c \in \mathbb{R}^m$. Four standard closure rules are used [2, 6]: if K_1, K_2 are PSD then (K1) $\lambda_1 K_1 + \lambda_2 K_2$ is PSD for $\lambda_1, \lambda_2 \geq 0$; (K2) the product $K_1 K_2$ is PSD, by the Schur product theorem; (K3) $K(x, y) = f(x)f(y)$ is PSD for any real f ; and (K4) a pointwise limit of PSD kernels is PSD.

The RBF kernel is PSD. The linear kernel is PSD because

$$c^\top G c = \sum_i \sum_j c_i c_j x_i^\top x_j = \left\| \sum_i c_i x_i \right\|_2^2 \geq 0, \quad (7.5)$$

so $(x^\top y)^k$ is PSD for every integer $k \geq 0$ by repeated application of (K2). The exponential series has non-negative coefficients, hence

$$\exp[2\gamma x^\top y] = \sum_{k=0}^{\infty} \frac{(2\gamma)^k}{k!} (x^\top y)^k \quad (7.6)$$

is a limit of non-negative combinations of PSD kernels and is PSD by (K1) and (K4). Expanding the squared norm as $\|x - y\|_2^2 = \|x\|_2^2 - 2x^\top y + \|y\|_2^2$ factors the RBF kernel into

$$\exp[-\gamma\|x - y\|_2^2] = \underbrace{e^{-\gamma\|x\|_2^2} e^{-\gamma\|y\|_2^2}}_{\text{PSD by (K3)}} \cdot \underbrace{\exp[2\gamma x^\top y]}_{\text{PSD}}, \quad (7.7)$$

which is PSD by (K2). The Gram matrix is therefore PSD for any dataset and any $\gamma > 0$.

An explicit feature map. The same expansion produces φ directly. Writing the multinomial expansion of $(x^\top y)^k$ over multi-indices $\nu \in \mathbb{N}_0^p$, with $|\nu| = \sum_j \nu_j$, $\nu! = \prod_j \nu_j!$ and $x^\nu = \prod_j x_j^{\nu_j}$, gives

$$\exp[2\gamma x^\top y] = \sum_{\nu \in \mathbb{N}_0^p} \frac{(2\gamma)^{|\nu|}}{\nu!} x^\nu y^\nu. \quad (7.8)$$

Reinstating the two Gaussian factors and splitting each term symmetrically yields $K(x, y) = \langle \varphi(x), \varphi(y) \rangle$ with components

$$\varphi_\nu(x) = e^{-\gamma\|x\|_2^2} \sqrt{\frac{(2\gamma)^{|\nu|}}{\nu!}} x^\nu, \quad \nu \in \mathbb{N}_0^p. \quad (7.9)$$

The index set is all multi-indices, so $\mathcal{H}$ is infinite-dimensional: φ contains every monomial in the p features, damped by $e^{-\gamma\|x\|_2^2}$ and weighted so the series converges.

Consequences. Three points follow. First, the exact feature map is infinite-dimensional and is therefore not represented as a finite explicit feature vector in this analysis; the scalar $K(x_i, x_j)$ instead evaluates the required inner product directly. Second, PSD of the Gram matrix makes the dual objective concave, so every local optimum is globally optimal and there are no spurious local maxima. PSD alone does not require the optimal dual coefficients to be

unique. Third, the primal in $\mathcal{H}$ is convex with affine constraints and admits the strictly feasible point $(w, b, \xi) = (0, 0, \mathbf{21})$, so Slater’s condition holds exactly as in Section 4.5. Although $\mathcal{H}$ is infinite-dimensional, the representer theorem restricts an optimum to the finite span of the training feature vectors, which connects this statement to the finite Gram-matrix dual; the primal–dual gap of 2.2×10^{-11} reported for the fitted RBF model is therefore a meaningful check that the solver reached a global optimum.

7.3 Joint Grid Search and Nested Validation

The RBF analysis uses the same log transformation, median imputation, standardization, stratified folds, and random seed as the linear analysis. The joint grid contains nine values of C from 0.01 to 100.00 and nine values of γ from 1.00×10^{-3} to 10.00, both at half-logarithmic spacing. Thus each inner search evaluates 81 parameter pairs by mean AUROC.

In every outer fold, preprocessing and the complete 81-point grid search are fitted using only the outer training data. Table 13 shows that the selected pairs vary substantially across folds. This instability is itself relevant: with only 114 observations, the apparently best nonlinear scale depends on which observations enter the training set.

Table 13: Nested cross-validation for joint RBF tuning. AUROC values and parameter displays use two decimals or two-decimal scientific notation.

Fold	C	γ	Inner train	Inner validation	Outer AUROC
1	3.16	1.00e-03	0.97	0.95	0.92
2	10.00	1.00e-03	0.96	0.96	0.94
3	0.32	0.03	0.96	0.94	0.98
4	1.00	3.16e-03	0.95	0.92	0.98
5	0.01	1.00e-03	0.97	0.96	0.88
Mean	—	—	0.962	0.946	0.942

The full-data grid search selected $C = 0.01$ and $\gamma = 1.00 \times 10^{-3}$. Here γ is an inverse squared-bandwidth parameter: smaller values give wider, smoother kernels. This γ is the lower boundary of the prespecified grid, indicating that the validation criterion favors the smoothest candidate rather than a strongly nonlinear boundary. At the selected pair, mean training AUROC is 0.96 and mean validation AUROC is 0.95, a gap of only 0.01. The induced-feature-space primal and dual objectives both round to 1.12, with full-precision duality gap 2.23×10^{-11} .

Table 14: Full-data RBF model-selection and optimization audit.

Quantity	Value
Selected C	0.01
Selected γ	1.00e-03
Mean training AUROC	0.96
Mean validation AUROC	0.95
Train–validation gap	0.01
Largest gap anywhere on grid	0.46
Mean nested outer AUROC	0.94
Support vectors	113
Observations with $\xi_i > 0$	111
Primal objective	1.12
Dual objective	1.12

7.4 Overfitting Diagnostic

The selected RBF model is not strongly overfit, but the grid contains a clear demonstration of the risk. Figure 10 compares validation AUROC with the train–validation gap. At $C = 0.01$ and $\gamma = 10.00$, training AUROC is 1.00 while validation AUROC falls to 0.54, giving a gap of 0.46. Across all values of C at $\gamma = 10.00$, training AUROC remains 1.00 but validation AUROC is only 0.54–0.59. The high- γ kernel has therefore memorized the training folds without learning a stable decision rule.

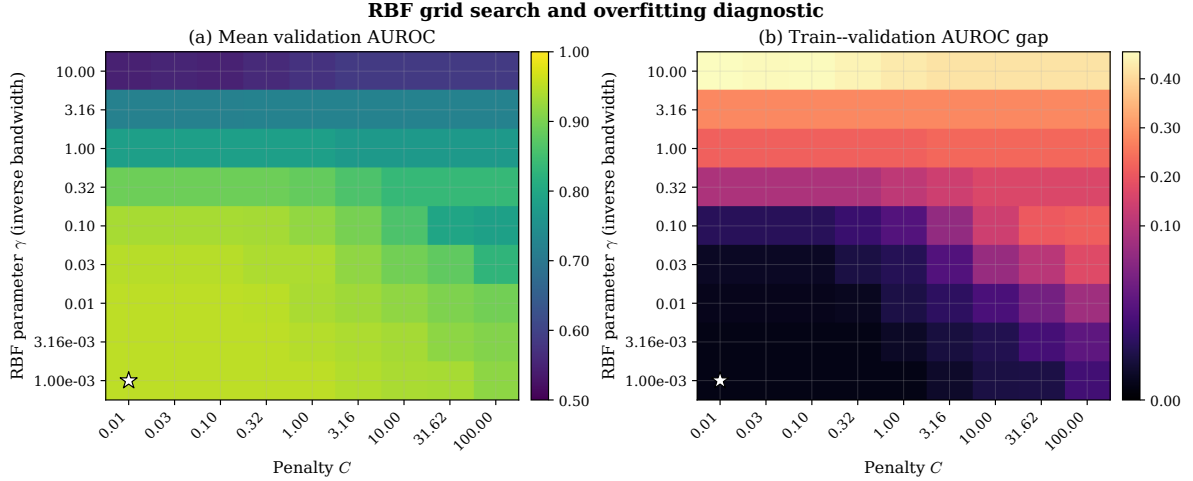


Figure 10: Joint RBF grid search. Panel (a) shows mean validation AUROC. Panel (b) shows mean training AUROC minus mean validation AUROC; large positive values indicate overfitting. The white star marks the selected full-data pair $C = 0.01$, $\gamma = 1.00 \times 10^{-3}$.

This result illustrates why training performance cannot be used to choose a kernel. A nonlinear model can achieve perfect training ranking precisely where its held-out ranking is close to chance. Nested validation, the train–validation gap, and the position of the selected point on the grid must be interpreted together.

Figure 11 shows the RBF decision surface in the same principal-component projection as the linear boundary of Figure 8. The kernel lets the boundary curve, but the two convex hulls still overlap: the added flexibility bends the surface around individual points without producing a clean separation in this illustrative two-dimensional fit. This picture motivates the comparison but does not decide which model generalizes better; that conclusion comes from the untouched outer folds below.

7.5 Pareto Frontier of Validation Performance and Overfitting

A Pareto analysis makes this trade-off explicit. The two objectives are to maximize mean validation AUROC and minimize the train–validation AUROC gap. Grid setting A dominates setting B when A has validation AUROC no smaller and an overfitting gap no larger, with at least one strict improvement. A setting is Pareto-optimal when no other setting dominates it. The hyperparameters C and γ identify the grid settings; they are not themselves the two Pareto objectives.

Figure 12 plots every evaluated setting. For the linear soft-margin grid, the frontier collapses to one point:

$$C = 0.01, \quad \text{AUROC}_{\text{val}} = 0.9452, \quad \text{gap} = 0.0141. \quad (7.10)$$

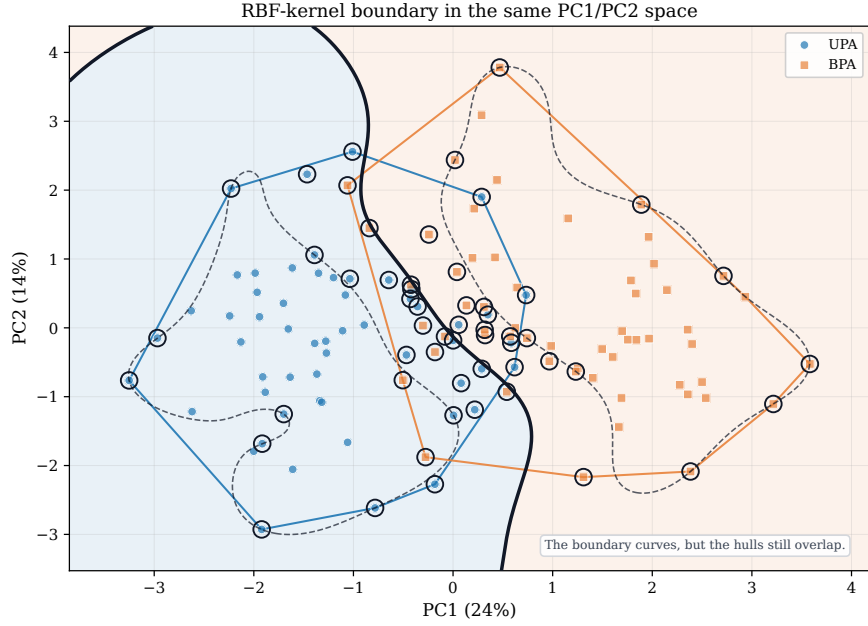


Figure 11: RBF-kernel decision boundary in the first two principal components, to be compared with the straight linear boundary of Figure 8. The boundary curves but the subtype hulls still overlap. Illustrative two-dimensional fit at a moderate γ ; the tuned model selected a much smoother γ .

Every other tested linear value of C has both a lower validation AUROC and a larger train-validation gap.

The RBF grid has a four-setting Pareto plateau. All four use $\gamma = 0.00316$ and

$$C \in \{0.01, 0.0316, 0.10, 0.3162\}, \quad (7.11)$$

with validation AUROC 0.9511 and gap 0.0083 to stored precision. Because their two Pareto outcomes are indistinguishable, the smallest value $C = 0.01$ is the natural simplicity tie-break. The ordinary grid search reported $\gamma = 0.001$ because it has the same maximum validation AUROC and appears first among tied scores; the Pareto diagnostic instead identifies $\gamma = 0.00316$ because its train-validation gap is smaller by approximately 2.06×10^{-6} . This difference is far below the fold-to-fold standard deviations. It is numerically negligible and does not constitute evidence of a meaningful nonlinear advantage.

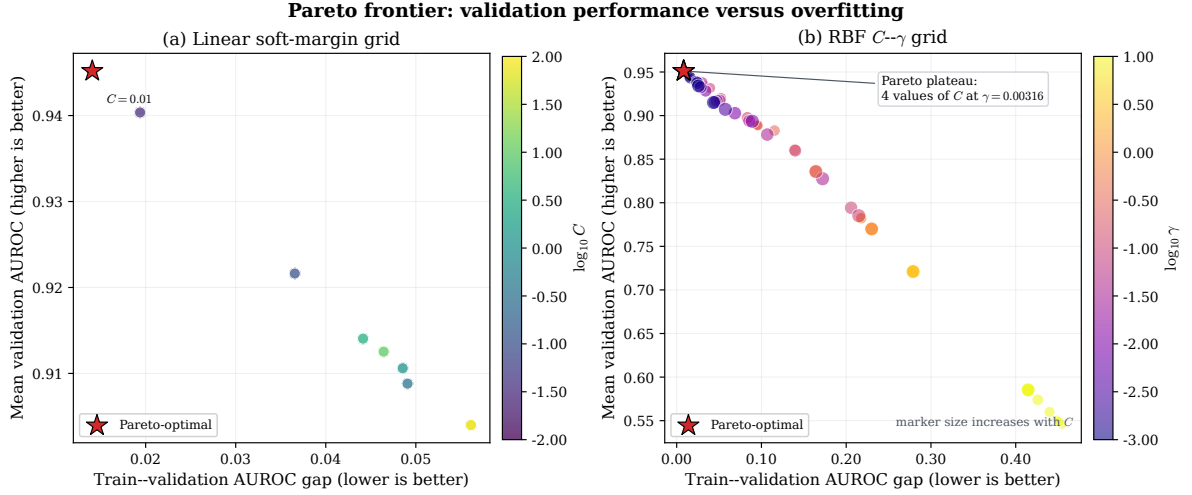


Figure 12: Pareto analysis of the linear and RBF tuning grids. The horizontal axis is the train-validation AUROC gap, so smaller values are preferred; the vertical axis is mean validation AUROC, so larger values are preferred. Red stars mark non-dominated settings. In panel (a), color represents $\log_{10} C$. In panel (b), color represents $\log_{10} \gamma$ and marker size increases with C .

This frontier is a tuning diagnostic computed from the cross-validated grid; it is not an independent performance estimate. Selecting a point from the frontier and then quoting the same validation folds would be optimistic. The final linear-versus-RBF conclusion therefore continues to use the untouched nested outer folds in the next subsection.

7.6 Comparison with the Linear Soft-Margin Model

For the prespecified seed-zero split, the comparison uses mean outer-fold AUROC because each outer fold is untouched by its corresponding parameter search. The linear model achieves 0.95 ± 0.04 , whereas the RBF model achieves 0.94 ± 0.04 . The RBF-minus-linear difference in mean outer AUROC is -3.17×10^{-3} , so there is no observed improvement from the kernel. The RBF accuracy is 0.82, sensitivity is 0.73, and specificity is 0.91, compared with approximately 0.86 for all three quantities under the linear model.

Table 15: Seed-zero nested cross-validation comparison of linear and RBF soft-margin SVMs. SD is the sample standard deviation across five outer folds (ddof = 1). Mean outer-fold AUROC avoids cross-fold score-scale artifacts; repeated-seed results provide the headline stability comparison.

Model	Mean outer AUC	SD	Pooled AUC	Accuracy	Sensitivity	Specificity
Linear soft margin	0.95	0.04	0.95	0.86	0.86	0.86
RBF kernel	0.94	0.04	0.85	0.82	0.73	0.91

The pooled RBF AUROC is shown for completeness but is not the primary estimate. Different outer folds selected different (C, γ) pairs, so their raw decision scores have different scales and pooling can alter cross-fold ranks. The mean of the five independently evaluated outer-fold AUROCs avoids this scale-comparability problem. This warning concerns rank pooling for AUROC; accuracy, sensitivity, and specificity use each classifier’s invariant decision-sign threshold and are not invalidated by positive score rescaling. Figure 13 therefore averages fold-specific ROC curves on a common false-positive-rate grid.

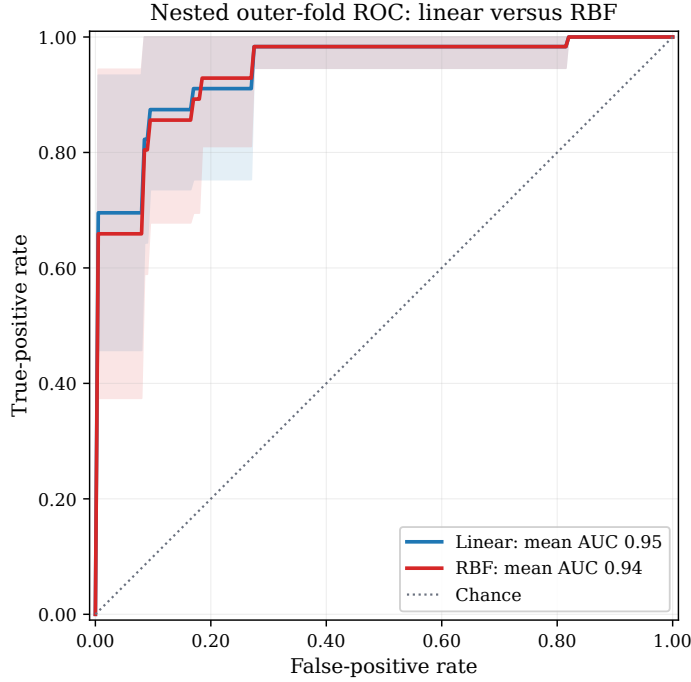


Figure 13: Mean nested outer-fold ROC curves for the linear and RBF soft-margin models. Shaded bands show one standard deviation across the five outer folds.

This seed-zero result is illustrative; the twenty-seed analysis in Section 9.1 is the more stable headline comparison. The present data do not support retaining the RBF kernel. It adds a second hyperparameter, produces unstable fold-specific choices, selects the smoothest available γ on the full dataset, and does not improve the mean nested outer-fold AUROC. With only 114 synthetic observations, a flexible high- γ boundary has a substantial overfitting risk. The linear soft-margin model is therefore the preferred specification unless a larger independent dataset demonstrates a reproducible nonlinear improvement.

The numerical values of C should not be compared literally across the linear and RBF models: kernel normalization changes the geometry and effective regularization scale. The comparison is between separately tuned model families using untouched outer folds, not between equal-valued penalties.

The complete grid, nested predictions, support-vector audit, primal–dual checks, comparison table, and figures are generated by `scripts/kernel_rbf_svm.py` and written under `output/csv/` and `output/figures/`.

8 Integrated Results Summary

The feasible ten-row calculation in Appendix A is an algebraic illustration only. The report’s feasibility conclusion is based on the complete standardized dataset and all five cross-validation training folds. Every full analysis problem was declared infeasible by the HiGHS linear-programming solver, as summarized earlier in Table 7.

These outcomes show that the UPA and BPA observations overlap in the selected ten-dimensional linear feature space. This is a property of the data geometry rather than a failure of the quadratic optimizer. Section 5.1 strengthens the solver status with an explicit Farkas certificate: a nonnegative vector satisfies $U^\top \lambda \approx 0$ and $\mathbf{1}^\top \lambda = 1$, proving that the standardized class convex hulls intersect. Consequently, the hard-margin primal cannot be fitted and no hard-margin ROC

exists.

The soft-margin model converts this empty feasible set into a feasible convex program by allowing penalized slack. Grid search selected $C = 0.01$. Its mean five-fold validation AUROC was 0.95, while seed-zero nested cross-validation produced pooled out-of-fold AUROC 0.95, accuracy 0.86, sensitivity 0.86, and specificity 0.86. The full-data primal and dual objectives both round to 0.63, with a full-precision duality gap of 4.85×10^{-10} . These soft-margin results are kept distinct from the infeasible hard-margin formulation.

On the illustrative seed-zero partition, the RBF experiment did not improve mean outer-fold AUROC (0.94 ± 0.04 versus 0.95 ± 0.04 for linear). Across twenty split seeds, however, both model families average approximately 0.94 (linear 0.9401, RBF 0.9399), with no stable winner. The selected full-data pair was $C = 0.01$ and $\gamma = 1.00 \times 10^{-3}$, with γ at the smoothest boundary of the grid. High- γ combinations achieved training AUROC 1.00 while falling as low as validation AUROC 0.54, demonstrating substantial overfitting. The Pareto audit retained one linear setting and a four-setting RBF plateau when validation AUROC was maximized jointly with minimization of the train-validation gap; it did not change the parsimony-based preference for the linear model after the repeated nested comparison found an effective tie.

9 Discussion and Limitations

The derivative analysis confirms that the hard-margin primal is a simple convex quadratic program: its objective has gradient $[w^\top, 0]^\top$ and positive-semidefinite Hessian $\text{diag}(I_p, 0)$, while each constraint is affine. These properties make any feasible instance globally tractable. They do not, however, guarantee that a feasible point exists.

The feasible teaching subset does not contradict the full-data result: feasibility is lost when the remaining 104 margin constraints are added.

The negative hard-margin feasibility result is substantively consistent with the application. Clinical measurements from UPA and BPA patients can overlap because of biological heterogeneity, measurement variation, and incomplete diagnostic information. A hard-margin linear model assumes away this overlap. The soft-margin formulation is therefore not a workaround for a failed solver; it is a different, explicitly stated optimization model whose slack variables quantify the overlap.

AUROC evaluates ranking over all thresholds; it does not establish calibrated probabilities or a clinically acceptable operating point. The threshold-zero sensitivity and specificity are descriptive for these fitted margins and were not chosen from a decision-analytic cost function. A real subtype workflow would also need comparison with definitive clinical procedures such as adrenal venous sampling, prospective calibration, and external validation. None of those claims can be made from this synthetic cohort.

The selected value $C = 0.01$ lies at the lower end of the prespecified grid. This indicates that stronger regularization was preferred to aggressively penalizing every violation. The neighboring value $C = 0.03$ produced a similar mean validation AUROC, so the exact numerical optimum should not be treated as a universal clinical constant. It is specific to this feature scaling, candidate grid, synthetic cohort, folds, and selection metric. Nested cross-validation reduces selection optimism but does not replace external validation.

Because both primary selections touched lower grid boundaries, an explicitly post-selection sensitivity analysis extended C to 10^{-4} and γ to 10^{-5} on the full-data five-fold validation split. Table 16 shows that the validation criterion continues toward an even more strongly regularized, smoother limit: four linear C values from 10^{-4} through 3.16×10^{-3} tie at AUROC 0.950, and the extended RBF maximum is attained on a three-setting plateau whose smallest setting is

$C = 10^{-4}, \gamma = 10^{-5}$ (AUROC 0.953). This exploratory audit does not replace or retroactively retune the prespecified nested-CV protocol. Instead, it shows that neither hyperparameter is precisely identified and that the data favor an almost linear, heavily regularized ranking. The tiny validation differences are well below fold-level uncertainty.

Table 16: Post-selection lower-bound sensitivity check. The primary choices remain the prespecified-grid results used in nested validation.

Model	Primary grid choice	Extended-grid result	Interpretation
Linear	$C = 0.01$	best AUROC 0.950; $C \in [10^{-4}, 3.16 \times 10^{-3}]$ tied	lower- C plateau
RBF	$C = 0.01, \gamma = 10^{-3}$	best AUROC 0.953; 3-setting tie; smallest $C = 10^{-4}, \gamma = 10^{-5}$	smoother-limit solution

The RBF comparison reinforces this limitation. Jointly searching 81 (C, γ) combinations increases the opportunity to select noise, and the preferred pair varied substantially between outer folds. The nested analysis found no performance gain, while the high- γ region produced perfect training AUROC and near-chance validation. Model complexity should therefore not be justified by training fit.

9.1 Robustness of the Linear–RBF Comparison

The primary comparison above uses mean AUROC across untouched outer folds. Two additional diagnostics explain why the conclusion is deliberately modest: the evidence supports choosing the simpler linear model, but it does not show that the linear model has intrinsically superior discrimination.

Pooled out-of-fold discrimination. On the pooled out-of-fold decision scores, the linear soft-margin AUROC (0.947) exceeds the RBF AUROC (0.851). This number is not a valid primary head-to-head estimate for the score-scale reason established in Section 7.6. The pooled gap and its naive DeLong or patient-bootstrap significance tests are therefore not used to choose the model, and no formal inference is attached to this scale-sensitive pooled calculation.

Mean outer-fold discrimination and seed robustness. The fairer comparison computes AUROC within each untouched outer fold and then averages the five values. Repeating the full nested procedure under twenty fold-assignment seeds, the linear and RBF estimates track each other closely (Figure 14). The mean difference is $+0.0002$, the descriptive 95% interval across seeds is $[-0.022, +0.017]$, and the linear model equals or exceeds the RBF model in twelve of twenty runs. These repeated splits reuse the same 114 observations and are therefore a stability analysis, not twenty independent experiments; the displayed empirical quantiles have no frequentist coverage interpretation. They show that the direction of the tiny gap depends on the split and provide no stable discrimination advantage for either model.

Interpretation. The diagnostics answer different questions. The pooled calculation exposes a cross-fold score-scaling problem; the repeated nested analysis shows that the proper fold-wise comparison is essentially tied. Because nonlinear flexibility does not deliver a stable outer-fold gain, the linear model is preferred on grounds of parsimony, fewer tuning degrees of freedom, and a directly inspectable standardized-space score.

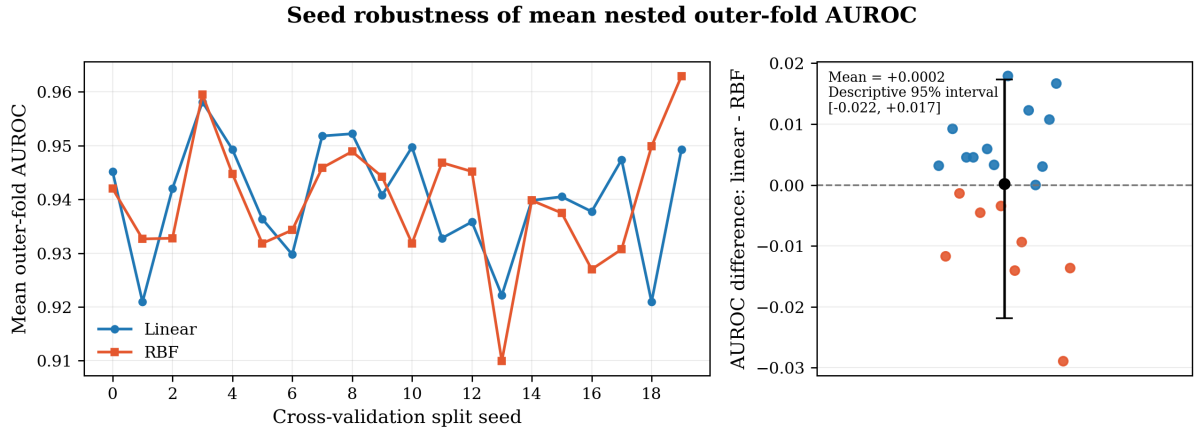


Figure 14: Seed-robustness of the fairer mean outer-fold AUROC. Panel (a): linear and RBF values across twenty fold-assignment seeds. Panel (b): the per-seed difference (linear minus RBF) with its mean and descriptive 95% interval. The interval straddles zero, showing that neither model has a stable within-fold discrimination advantage across splits.

Simple baselines. Table 17 places the SVM comparison in context using the same twenty outer-split seeds. A prespecified PAC-only ranking reaches mean AUROC 0.809, while a leakage-safe, nested-tuned L2 logistic regression reaches 0.944. The linear and RBF SVM means are both 0.940 to three decimals. Thus most of the simulated signal is multivariable, but the kernel provides no clear gain over conventional linear classifiers. The logistic baseline’s 0.004 advantage over the linear SVM is likewise not stable: it favors logistic in twelve of twenty splits, the same proportion by which the linear SVM leads the RBF. Across these four models, only the PAC-only baseline separates clearly from the rest.

Table 17: Baseline comparison across twenty split seeds. Seed SD and range describe split sensitivity on the same synthetic cohort; they are not independent-sample uncertainty intervals.

Model	Mean AUROC	Seed SD	Seed range
PAC only	0.809	0.008	0.797–0.829
L2 logistic	0.944	0.008	0.929–0.963
Linear SVM	0.940	0.011	0.921–0.958
RBF SVM	0.940	0.012	0.910–0.963

The analysis also uses synthetic rather than patient-level data. Its purpose is therefore methodological: it determines whether the requested optimization model is compatible with the supplied project dataset. It does not estimate clinical utility or external generalization.

10 Conclusion

This report first formulated PA subtype classification as a linear hard-margin SVM and showed, by solver status, Farkas’ lemma, and intersecting convex hulls, that the complete dataset and every training fold are nonseparable. The hard-margin primal therefore has no feasible solution and cannot produce a valid ROC curve. The subsequent soft-margin primal introduces slack variables, while its dual replaces the hard-margin constraint $\alpha_i \geq 0$ with the box constraint $0 \leq \alpha_i \leq C$. Leakage-safe grid search selected $C = 0.01$. The seed-zero nested mean outer-fold AUROC was 0.95, while the twenty-seed mean was 0.94. The optimized soft-margin primal and

dual values agree to numerical tolerance. The combined analysis then tested an RBF kernel by jointly tuning C and γ . The RBF seed-zero mean outer-fold AUROC was 0.94, and its twenty-seed mean was also 0.94; highly localized kernels showed severe train–validation gaps. The linear soft-margin SVM is therefore preferred for parsimony, not because the data demonstrate superior discrimination. This conclusion preserves the central distinctions: hard-margin infeasibility is a statement about data geometry, soft-margin optimization explicitly trades margin width against violations, and a kernel should be retained only when its nonlinear flexibility improves performance on data not used for tuning.

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A Worked Teaching Examples

A.1 Worked Separable Primal–Lagrangian–Dual Example

This deliberately small example makes the primal, Lagrangian, and dual views transparent. It uses the five observations in Table 18. Each point is embedded in the same ten-dimensional notation as the main analysis by setting $x_3 = \dots = x_{10} = 0$. The symbol y_i used in this example has exactly the same role as t_i elsewhere in the report. Exact integers are retained where they make the symbolic constraint algebra easier to read.

Table 18: Five-point teaching dataset embedded in ten feature dimensions.

Point	x_1	x_2	$x_3, \dots, x_{10}$	y
A	−2.00	0.00	0.00	−1
B	−1.00	1.00	0.00	−1
C	1.00	1.00	0.00	+1
D	2.00	0.00	0.00	+1
E	2.00	2.00	0.00	+1

Geometric intuition. The closest opposite-class observations in the horizontal direction are B at $x_1 = -1.00$ and C at $x_1 = 1.00$. The widest symmetric separating strip therefore places its decision boundary halfway between them at $x_1 = 0.00$, with supporting margins at $x_1 = -1.00$ and $x_1 = 1.00$. Figure 15 shows this geometry. B and C touch the margins, while A, D, and E are farther from the decision boundary.

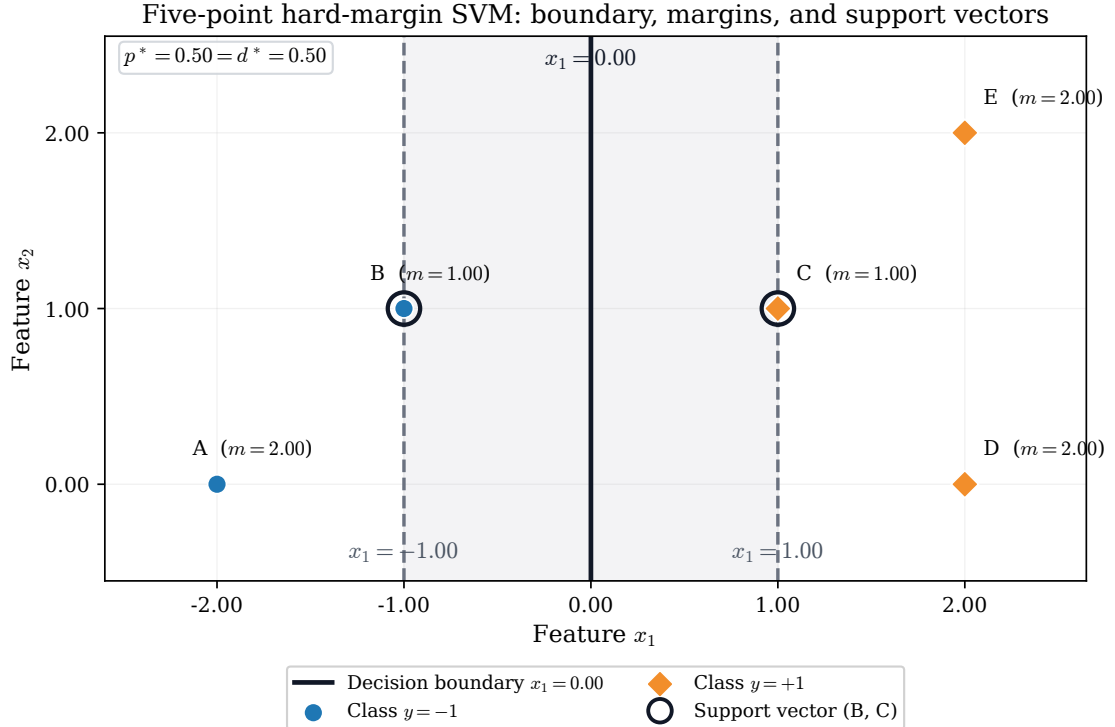


Figure 15: Five-point hard-margin example. The black line is the decision boundary, the dashed lines are the two supporting margins, and the rings identify support vectors B and C. Here $m_i = y_i(w^\top x_i + b)$ denotes the signed functional margin.

Primal problem and the five constraints. For $w = (w_1, \dots, w_{10})^\top$, the hard-margin primal is

$$\begin{aligned} \min_{w, b} \quad & \frac{1}{2} \|w\|_2^2 = \frac{1}{2} \sum_{j=1}^{10} w_j^2 \\ \text{subject to} \quad & y_i(w^\top x_i + b) \geq 1, \quad i \in \{A, B, C, D, E\}. \end{aligned} \quad (\text{A.1})$$

Substituting every observation gives

$$\begin{aligned} \text{A:} \quad & -1[-2w_1 + 0w_2 + 0w_3 + \dots + 0w_{10} + b] \geq 1 \iff 2w_1 - b \geq 1, \\ \text{B:} \quad & -1[-w_1 + w_2 + 0w_3 + \dots + 0w_{10} + b] \geq 1 \iff w_1 - w_2 - b \geq 1, \\ \text{C:} \quad & +1[w_1 + w_2 + 0w_3 + \dots + 0w_{10} + b] \geq 1 \iff w_1 + w_2 + b \geq 1, \\ \text{D:} \quad & +1[2w_1 + 0w_2 + 0w_3 + \dots + 0w_{10} + b] \geq 1 \iff 2w_1 + b \geq 1, \\ \text{E:} \quad & +1[2w_1 + 2w_2 + 0w_3 + \dots + 0w_{10} + b] \geq 1 \iff 2w_1 + 2w_2 + b \geq 1. \end{aligned} \quad (\text{A.2})$$

This is not an unconstrained optimization problem. If one differentiates only the objective and sets its gradient to zero, then

$$\nabla_w \left(\frac{1}{2} \|w\|_2^2 \right) = w = 0. \quad (\text{A.3})$$

At $w = 0$, the positive-class constraints require $b \geq 1$, whereas the negative-class constraints require $b \leq -1$; no intercept can satisfy both. The unconstrained minimum is therefore infeasible, which is precisely why the margin constraints must enter the optimality analysis through the Lagrangian and KKT conditions.

Solving the primal step by step. Add the simplified constraints for B and C:

$$(w_1 - w_2 - b) + (w_1 + w_2 + b) \geq 2 \implies w_1 \geq 1. \quad (\text{A.4})$$

It follows that every feasible point satisfies

$$\frac{1}{2} \|w\|_2^2 = \frac{1}{2} (w_1^2 + w_2^2 + \dots + w_{10}^2) \geq \frac{1}{2}. \quad (\text{A.5})$$

Choose $w_1 = 1.00$, $w_2 = \dots = w_{10} = 0.00$. The B constraint then gives $1.00 - b \geq 1.00$, hence $b \leq 0.00$, while the C constraint gives $1.00 + b \geq 1.00$, hence $b \geq 0.00$. Thus $b = 0.00$. This candidate also satisfies the A, D, and E constraints, so it attains the lower bound and is globally optimal:

$$w^* = (1.00, 0.00, \dots, 0.00)^\top, \quad b^* = 0.00, \quad p^* = \frac{1}{2} \|w^*\|_2^2 = 0.50. \quad (\text{A.6})$$

The decision equation $(w^*)^\top x + b^* = 0$ reduces to $x_1 = 0.00$.

Table 19: Signed functional margins at the primal optimum.

Point	y_i	$(w^*)^\top x_i + b^*$	$y_i((w^*)^\top x_i + b^*)$	Status
A	-1	-2.00	2.00	Outside margin
B	-1	-1.00	1.00	Support vector
C	+1	1.00	1.00	Support vector
D	+1	2.00	2.00	Outside margin
E	+1	2.00	2.00	Outside margin

Table 19 confirms that B and C are support vectors because their signed margins equal 1.00. A, D, and E have signed margins strictly greater than 1.00 and therefore lie outside the margin strip.

Lagrangian, stationarity, and complementary slackness. Introduce one multiplier $\alpha_i \geq 0$ for each of the five constraints:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|_2^2 - \sum_{i \in \{A, B, C, D, E\}} \alpha_i \left[y_i (w^\top x_i + b) - 1 \right]. \quad (\text{A.7})$$

Stationarity with respect to the primal variables gives

$$\begin{aligned} \nabla_w \mathcal{L} = 0 &\implies w = \sum_i \alpha_i y_i x_i, \\ \frac{\partial \mathcal{L}}{\partial b} = 0 &\implies \sum_i \alpha_i y_i = 0. \end{aligned} \quad (\text{A.8})$$

Complementary slackness requires

$$\alpha_i \left[y_i (w^\top x_i + b) - 1 \right] = 0 \quad \text{for every } i. \quad (\text{A.9})$$

For A, D, and E the bracket equals $2.00 - 1.00 = 1.00 > 0$, so complementary slackness forces

$$\alpha_A = \alpha_D = \alpha_E = 0.00. \quad (\text{A.10})$$

For B and C the bracket is zero, so their multipliers may be positive. The intercept stationarity equation becomes

$$-\alpha_B + \alpha_C = 0 \implies \alpha_B = \alpha_C = \gamma. \quad (\text{A.11})$$

Substituting this equality into the weight stationarity equation gives

$$\begin{aligned} w &= \gamma(-1)(-1, 1, 0, \dots, 0) + \gamma(+1)(1, 1, 0, \dots, 0) \\ &= (2\gamma, 0, \dots, 0). \end{aligned} \quad (\text{A.12})$$

Matching the primal solution $w^* = (1.00, 0.00, \dots, 0.00)$ gives $2.00\gamma = 1.00$, and therefore

$$\alpha_B = \alpha_C = 0.50. \quad (\text{A.13})$$

In the two visible coordinates, the reconstruction is

$$0.50(-1)(-1.00, 1.00) + 0.50(+1)(1.00, 1.00) = (1.00, 0.00), \quad (\text{A.14})$$

which verifies that the two support vectors determine the separating direction.

Dual problem and strong duality. Minimizing the Lagrangian over w and b using the stationarity equations produces the hard-margin dual

$$\begin{aligned} \max_{\alpha} \quad q(\alpha) &= \sum_i \alpha_i - \frac{1}{2} \left\| \sum_i \alpha_i y_i x_i \right\|_2^2 \\ \text{subject to} \quad \alpha_i &\geq 0, \quad \sum_i \alpha_i y_i = 0. \end{aligned} \quad (\text{A.15})$$

For this dataset, the two nonzero coordinates of $\sum_i \alpha_i y_i x_i$ are

$$\begin{aligned} z_1 &= 2\alpha_A + \alpha_B + \alpha_C + 2\alpha_D + 2\alpha_E, \\ z_2 &= -\alpha_B + \alpha_C + 2\alpha_E, \end{aligned} \quad (\text{A.16})$$

so the complete dataset-specific dual is

$$\begin{aligned} \max_{\alpha \geq 0} \quad & \alpha_A + \alpha_B + \alpha_C + \alpha_D + \alpha_E \\ & - \frac{1}{2} \left[(2\alpha_A + \alpha_B + \alpha_C + 2\alpha_D + 2\alpha_E)^2 + (-\alpha_B + \alpha_C + 2\alpha_E)^2 \right] \\ \text{subject to} \quad & -\alpha_A - \alpha_B + \alpha_C + \alpha_D + \alpha_E = 0. \end{aligned} \quad (\text{A.17})$$

At $(\alpha_A, \alpha_B, \alpha_C, \alpha_D, \alpha_E) = (0.00, 0.50, 0.50, 0.00, 0.00)$, the equality constraint holds and

$$q(\alpha) = 0.50 + 0.50 - \frac{1}{2} [(0.50 + 0.50)^2 + (-0.50 + 0.50)^2] = 0.50. \quad (\text{A.18})$$

The primal construction already supplies a feasible value $p^* = 0.50$, while this dual-feasible point supplies the lower bound $d^* = 0.50$. Weak duality prevents a dual value from exceeding a primal value, so equality certifies that both points are optimal:

$$\boxed{d^* = 0.50 = p^*}. \quad (\text{A.19})$$

This zero duality gap is the numerical manifestation of strong duality. The figure and all calculations in this subsection are reproduced by `scripts/build_svm_primal_dual_toy_example.py`.

A.2 Ten-Observation Spreadsheet Illustration

To make the matrix constraint concrete, a self-contained teaching table was constructed from the first ten observations in the project spreadsheet. The selection rule is transparent: scan from top to bottom, take the first five UPA and first five BPA observations, and restore source-row order. In this dataset these are exactly rows 2–11. The four hormone variables are log-transformed and all ten features are standardized using statistics from these ten rows. No selected value is missing, so median imputation is inactive.

For $p = 10$, the vectors used in one constraint row are

$$\theta = \begin{bmatrix} w_1 & \cdots & w_{10} & b \end{bmatrix}^\top, \quad u_i = t_i \begin{bmatrix} x_{i1} & \cdots & x_{i,10} & 1 \end{bmatrix}^\top. \quad (\text{A.20})$$

Their inner product is the signed functional margin:

$$u_i^\top \theta = t_i (x_i^\top w + b) = t_i s_i. \quad (\text{A.21})$$

The expression is therefore not “ $u_i > 1$ ”: u_i is a vector. The scalar dot product $u_i^\top \theta$ must satisfy $u_i^\top \theta \geq 1$. For UPA, $t_i = +1$ and the rule becomes $s_i \geq 1$; for BPA, $t_i = -1$ and it becomes $s_i \leq -1$.

Solving the primal quadratic program for this teaching subsystem gives

$$\begin{aligned} w_{\text{teach}}^\top &\approx (0.34, -0.42, -0.65, 0.08, -0.10, \\ &\quad 0.16, -0.13, 0.39, -0.75, 0.64), \\ b_{\text{teach}} &\approx 0.29. \end{aligned} \quad (\text{A.22})$$

Numerical evaluation of the objective. The ten observations contribute ten margin constraints; they do not create ten additive loss terms. Thus the fitted teaching subsystem has

one global primal objective. Substituting the fitted weights gives

$$\begin{aligned}
\|w_{\text{teach}}\|_2^2 &\approx (0.34)^2 + (-0.42)^2 + (-0.65)^2 + (0.08)^2 + (-0.10)^2 \\
&\quad + (0.16)^2 + (-0.13)^2 + (0.39)^2 + (-0.75)^2 + (0.64)^2 \\
&\approx 1.90 \quad \text{from displayed coefficients,} \\
&= 1.88 \quad \text{from full-precision coefficients,} \\
\phi(\theta_{\text{teach}}) &= \frac{1}{2} \|w_{\text{teach}}\|_2^2 \\
&= \frac{1}{2} (1.88) \\
&\approx \boxed{0.94}.
\end{aligned} \tag{A.23}$$

The intercept $b_{\text{teach}} \approx 0.29$ is not squared because the hard-margin primal does not penalize the intercept. The ten rowwise quantities governed by this solution are the constraints $u_i^\top \theta \geq 1$ in Table 20.

The coefficient order is PAC, PRA, potassium, tumor size, 18-OHF, 18-oxoF, systolic blood pressure, diastolic blood pressure, DDD, and age. Table 20 shows every constraint. The complete raw x_i , transformed x_i , standardized x_i , u_i , parameter, gradient, and Hessian tables are supplied in the companion CSV folder `output/csv/hard_margin_10row/`.

Table 20: Teaching-only row-by-row evaluation of $u_i^\top \theta \geq 1$. Displayed values are rounded; feasibility uses unrounded values.

Patient	Class	t_i	Score s_i	$u_i^\top \theta$	$g_i = 1 - u_i^\top \theta$	Feasible	SV
CGH-001	UPA	+1	3.22	3.22	-2.22	Yes	No
CGH-002	UPA	+1	1.39	1.39	-0.39	Yes	No
CGH-003	UPA	+1	1.00	1.00	0.00	Yes	Yes
CGH-004	BPA	-1	-1.00	1.00	0.00	Yes	Yes
CGH-005	BPA	-1	-1.41	1.41	-0.41	Yes	No
CGH-006	BPA	-1	-1.59	1.59	-0.59	Yes	No
CGH-007	BPA	-1	-1.00	1.00	0.00	Yes	Yes
CGH-008	BPA	-1	-1.00	1.00	0.00	Yes	Yes
CGH-009	UPA	+1	2.27	2.27	-1.27	Yes	No
CGH-010	UPA	+1	1.00	1.00	0.00	Yes	Yes

For example, CGH-004 is a BPA observation, so $t_4 = -1$. Its score is $s_4 = -1$, giving $u_4^\top \theta = (-1)(-1) = 1$ and $g_4 = 0$. It lies exactly on the negative supporting margin and is a support vector. The corresponding spreadsheet calculation is

$$\begin{aligned}
s_i &= \text{SUMPRODUCT}(x_i, w) + b, \\
u_i^\top \theta &= t_i s_i, \\
\text{constraint check} &= (u_i^\top \theta \geq 1 - 10^{-8}).
\end{aligned} \tag{A.24}$$

Figure 16 plots the raw scores rather than the signed margins. This preserves the two class regions: BPA observations lie at or left of -1 , UPA observations lie at or right of $+1$, and $s = 0$ is the classifier boundary.

This feasible subsystem does not imply that the complete cohort is separable. Removing 104 constraints can leave a feasible problem even when adding the remaining constraints makes the full system infeasible. No ROC, AUROC, sensitivity, or specificity is calculated from this ten-row in-sample illustration.

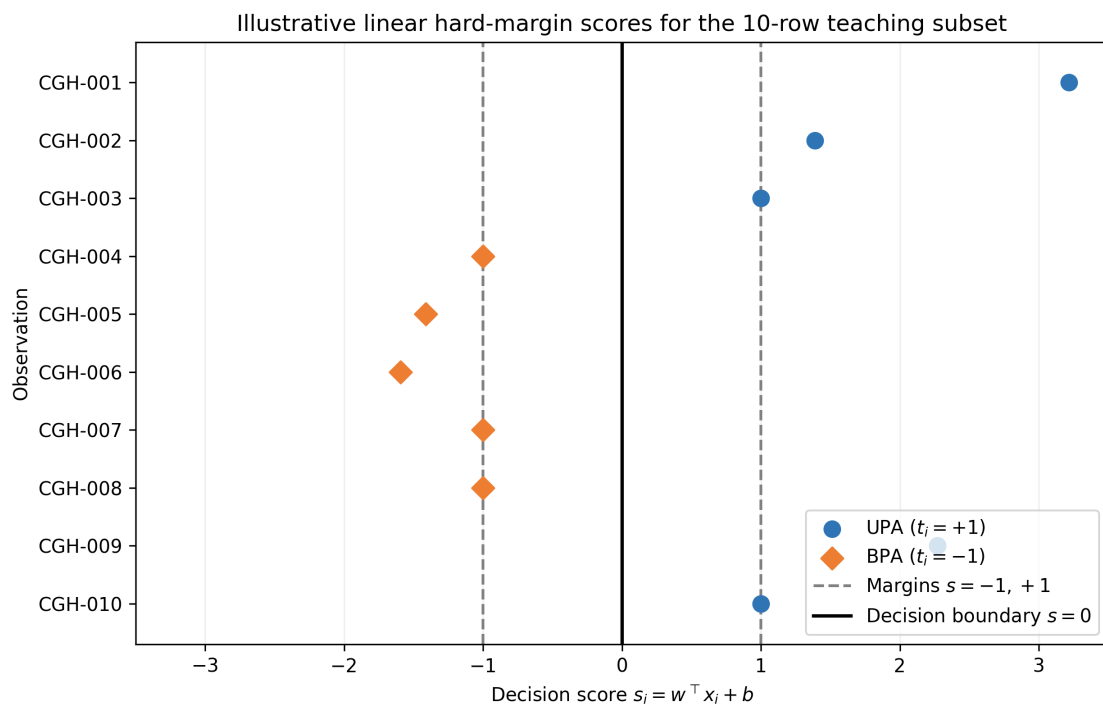


Figure 16: Decision scores for the separable ten-row teaching subsystem. This is an algebraic illustration, not a validation or ROC result.

B Internal Reproduction and Numerical Audit

The supplied numerical artifacts were regenerated and audited against the project scripts and dataset in the local reference environment (Python 3.14, scikit-learn 1.8, SciPy 1.17, NumPy 2.4, pandas 3.0, Matplotlib 3.10, and openpyxl 3.1). The corresponding package versions are recorded in `requirements.txt`. Generated full-precision CSV values were compared against the values reported in the main text.

- **Result tables.** The generated CSV artifacts reproduce the reported numerical results. Exact machine-zero residual mantissas can vary with summation order and the linear-algebra backend; values below 10^{-15} are therefore reported as tolerance statements rather than as reproducible decimal identities.
- **LaTeX tables.** The displayed table fragments are generated directly from their corresponding full-precision CSV artifacts.
- **Figure provenance.** Numerical figures generated by the documented scripts are reproducible from the shipped dataset and CSVs. Seven explanatory figures are committed static teaching assets: `feature_distributions`, `farkas_lemma_schematic`, `convex_hull_overlap_2d`, `fig_step3_c_grid`, `fig_step3_pareto`, `fig_step3c_pc_boundary`, and `fig_step4_rbf_pc_boundary`. They illustrate the same shipped calculations but are not claimed as script-generated outputs.
- **Headline metrics.** The linear soft-margin nested seed-zero out-of-fold AUROC (0.9474), accuracy (0.8596), sensitivity (0.8571), and specificity (0.8621), and the RBF selection $(C, \gamma) = (0.01, 10^{-3})$ with mean outer AUROC 0.9420, all reproduce to six decimal places within the supplied project pipeline. Across twenty fold-assignment seeds, the more stable headline means are 0.9401 for the linear SVM and 0.9399 for the RBF SVM.
- **Optimization quality.** The primal–dual gaps of both fitted models are small (4.9×10^{-10} for the linear model and 2.2×10^{-11} for the RBF model), confirming both quadratic programs are solved to the stated numerical tolerance.
- **Leakage safety.** The log transformation, median imputation, and standardization are all contained inside the cross-validation pipeline, so every preprocessing parameter is estimated from the training fold alone. The nested cross-validation is therefore free of information leakage.

The primary conclusion is confirmed: the RBF kernel did not provide a stable improvement over the linear soft-margin model. With only 114 observations and substantial overlap among repeated splits, the fold and seed repetitions are treated as validation and stability diagnostics rather than independent replicates for formal significance testing. Preferring the simpler linear model is therefore the supported, appropriately cautious conclusion.

C Source Code

All code, the synthetic dataset, and the generated figures and tables are available in the project repository:

https://github.com/nttssv/convex_optimization_svm_example

The repository is organized as follows. Each script is reached by appending its path to `https://github.com/nttssv/convex_optimization_svm_example/blob/main/`.

- `scripts/linear_hard_margin_svm.py` — fold-wise preprocessing, the hard-margin feasibility test by linear programming, and the constrained primal solve used in Section 4.
- `scripts/linear_soft_margin_svm.py` — leakage-safe nested cross-validation, selection of C by AUROC, the final soft-margin fit with its primal–dual audit, and the tables and figures of Section 6.
- `scripts/compare_primal_optimizers.py` — controlled comparison of stochastic subgradient, Heavy Ball momentum-subgradient, and Nesterov-style momentum-subgradient methods in Section 6.7.
- `scripts/kernel_rbf_svm.py` — joint C, γ tuning inside nested cross-validation, the train–validation gap diagnostic, and the RBF-versus-linear comparison of Section 7.
- `scripts/build_full_data_infeasibility_certificate.py` — the Farkas infeasibility certificate of Section 5.1.
- `scripts/build_descriptive_tables.py` — the descriptive statistics and class-contrast tables of Section 3.
- `scripts/build_svm_primal_dual_toy_example.py` — the five-point primal–dual teaching example in Appendix A.1.
- `scripts/build_hard_margin_teaching_workbook.py` — the ten-observation teaching subsystem and companion spreadsheet.
- `scripts/build_seed_robustness.py` — the repeated-seed stability diagnostic comparing the linear and RBF models.
- `scripts/build_boundary_sensitivity.py` — the explicitly post-selection audit of the lower C and γ grid boundaries.
- `scripts/build_baseline_comparison.py` — twenty-seed PAC-only, L2-logistic, linear-SVM, and RBF-SVM context comparison.
- `scripts/make_synthetic_dataset.py` — deterministic generation of the synthetic input cohort from seed 20260710.
- `scripts/_pubstyle.py` — shared plotting style used by the publication figures.
- Legacy exploratory utilities (unused by every headline result, table, and figure): `scripts/svm_convergence.py`; `scripts/svm_cv_eval.py`; and `scripts/svm_diagnostics_from_data.py`.
- `data/cgh_pa_dataset.csv` — the synthetic 114-patient cohort; `output/` — the regenerated figures, CSV audits, and $\text{\LaTeX}$ table fragments.

The exact execution order and PDF-build command are documented in `README.md`. Running those commands from the repository root, with `scripts/` on `PYTHONPATH`, regenerates the report’s tables, numerical audit files, and script-produced figures. The seven static teaching figures listed above are shipped directly in `output/figures/`.